



Scientific Committee on Consumer Safety

SCCS

SCIENTIFIC ADVICE ON

hair dye 'Basic blue 99'

(C059)

(CAS No. 68123-13-7, EC No. 268-544-3)

Submissions IV & V



The SCCS adopted this document on 2 February 2026

ACKNOWLEDGMENTS

SCCS members listed below are acknowledged for their valuable contribution to the finalisation of this scientific advice.

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SCCS

The Committee shall provide Opinions on questions concerning health and safety risks (notably chemical, biological, mechanical and other physical risks) of non-food consumer products (for example cosmetic products and their ingredients, toys, textiles, clothing, personal care and household products such as detergents, etc.) and services (for example: tattooing, artificial tanning, etc.).

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1. ABSTRACT

The SCCS concludes the following:

- (1) *In light of the data provided, does the SCCS consider Basic Blue 99 safe, when used as an ingredient at 1 % in non-oxidative hair dye formulations?*

Having evaluated all the available data, the SCCS does not consider the use of Basic Blue 99 safe for use as an ingredient in non-oxidative hair dye formulations because the collective evidence indicates a potential for genotoxicity.

- (2) *Does the SCCS have any further scientific concerns with regard to the use of Basic Blue 99 in cosmetic products?*

Although there is an uncertainty regarding the exact composition of Basic Blue 99 batch that caused (partly severe) immediate-type hypersensitivity reactions, the SCCS considers this an alert for such a risk with the current batch specifications under consideration. The SCCS regards Basic Blue 99 a skin sensitiser.

Keywords: SCCS, scientific advice, hair dye, Basic Blue 99, C059, 3-[(4-amino-6-bromo-5,8-dihydro-1-hydroxy-8-imino-5-oxo-2-naphtyl)amino]-N,N,N-trimethylanilinium chloride, CAS No. 68123-13-7, EC No. 268-544-3, submission IV & V, Regulation 1223/2009

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Table of Content

1.	ABSTRACT	4
2.	MANDATE FROM THE EUROPEAN COMMISSION	6
3.	SCIENTIFIC ADVICE	7
3.1	Physicochemical Specifications	7
3.1.1	Chemical identity	7
3.1.2	Physical form	10
3.1.3	Molecular weight	10
3.1.4	Purity, composition and substance codes	11
3.1.5	Impurities / accompanying contaminants	15
3.1.6	Solubility	25
3.1.7	Partition coefficient (Log P _{ow})	25
3.1.8	Additional physicochemical specifications	26
3.1.9	Homogeneity and Stability	26
3.2	Function and uses	27
3.3	Toxicological evaluation	27
3.3.1.	Acute toxicity	27
3.3.2.	Irritation and corrosivity	28
3.3.3.	Skin sensitisation	30
3.3.4.	Dermal / percutaneous absorption	31
3.3.5.	Repeated dose toxicity	32
3.3.6.	Mutagenicity / Genotoxicity	34
3.3.7.	Carcinogenicity	47
3.3.8.	Reproductive toxicity	47
3.3.9.	Toxicokinetics	48
3.3.10.	Photo-induced toxicity	48
3.3.11.	Human data	49
3.3.12.	Special investigations	49
3.3.13.	Safety evaluation (including calculation of the MoS)	49
3.3.14.	Discussion	50
4.	CONCLUSION	52
5.	MINORITY OPINION	52
6.	REFERENCES	53
7.	GLOSSARY OF TERMS	56
8.	LIST OF ABBREVIATIONS	56

2. MANDATE FROM THE EUROPEAN COMMISSION

Background

The ingredient with the INCI name hair dye 'Basic Blue 99' and chemical name '3-[(4-amino-6-bromo-5,8-dihydro-1-hydroxy-8-imino-5-oxo-2-naphtyl)amino]-N,N,N-trimethylanilinium chloride' (CAS No. 68123-13-7, EC No. 268-544-3) may be used as an ingredient in non-oxidative hair dye formulations.

In September 2011, the Scientific Committee for Consumer Safety (SCCS) concluded that: 'Basic Blue 99 is a mixture of 23-32 substances of varying concentrations as demonstrated by the HPLC analysis of two batches RS2798801 and 74/75'. The SCCS noted that due to the highly variable composition of Basic Blue 99 in these batches, its safety could not be evaluated (SCCS/1437/11)¹.

In July 2014, the applicant provided new analytical data (Submission III) on the batches used in the previous submissions and on other more recent batches. The SCCS conclude, in September 2014, that 'Basic Blue 99 is a mixture of up to 40 substances of varying concentrations as demonstrated by the HPLC analysis of six' and reiterated their position that due to the highly variable composition of Basic Blue 99 in six batches, its safety could not be evaluated (SCCS/1537/14)².

With the current submission received in February 2025, the applicant requests to assess the safety of Basic Blue 99 in view of the newly provided information on composition and analytical description, as well as additional genotoxicity results, when Basic Blue 99 is used as an ingredient at 1 % in non-oxidative hair dye formulations.

Terms of reference

(1) In light of the data provided, does the SCCS consider Basic Blue 99 safe, when used as an ingredient at 1 % in non-oxidative hair dye formulations?

(2) Does the SCCS have any further scientific concerns with regard to the use of Basic Blue 99 in cosmetic products?

¹ https://ec.europa.eu/health/scientific_committees/consumer_safety/docs/sccs_o_068.pdf

² https://ec.europa.eu/health/scientific_committees/consumer_safety/docs/sccs_o_161.pdf

3. SCIENTIFIC ADVICE

Preamble

In April 2016, EFfCI submitted another dossier (Submission IV) containing new information on composition in an update of the analytical description of market quality and other data. In June 2017, the SCCS could not conclude on the safety of Basic Blue 99 (C059) because it is composed of several substances and isomeric forms, with a large variability between the composition of different batches. Also, the toxicological data provided in the previous submission do not relate to the material specifications provided for the assessment. The safety assessment of Basic Blue 99 will require a clear well-defined set of specifications for the composition of the material intended to be used in cosmetic products as well as supporting toxicological data relating to a representative composition (SCCS/1585/17).

The Applicant has provided new data in the current submission to support safety of Basic Blue 99 - especially in regard to purity/impurity profiles of different batches produced by improved production process.

3.1 Physicochemical Specifications

3.1.1 Chemical identity

3.1.1.1. Primary name and/or INCI name

Basic Blue 99 (INCI)

3.1.1.2 Chemical names

New data

Main component:

IUPAC name: 3-[(4-amino-6-bromo-5,8-dihydro-1-hydroxy-8-imino-5-oxo-2-naphthyl)amino]-N,N,N-trimethylanilinium chloride,

Other name: Benzenaminium, 3-[(4-amino-6-bromo-5, 8-dihydro-1-hydroxy-8-imino-5-oxo-2-naphthalenyl)amino]-N,N,N-trimethyl-, chloride (9CI)

According to the Applicant, the above reported chemical names correspond to the most abundant molecule in the substance = main component

Other major constituents of the substance:

IUPAC names:

1. 3-((8-amino-7-bromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride
2. 3-((8-amino-3,7-dibromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride
3. 3-((8-amino-3,7-dibromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride
4. 3-((8-amino-6,7-dibromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride

Ref: Submission V to SCCS- Basic Blue 99 -C059_- final

3.1.1.3 Trade names and abbreviations

New data

COLIPA C 059
Arianor Steel Blue
Jarocol Steel Blue
Basic Blue 99
C.I. 56059
CI Basic Blue 99
Hair dye C059

3.1.1.4 CAS / EC number

CAS: 68123-13-7
EC: 268-544-3

3.1.1.5 Structural formula

New data

According to the Applicant, the most abundant constituents of Basic Blue 99 (isolated from batch VW080224A, actual batch manufactured according to improved synthesis procedure and representative market materials), are mono- and dibromated isomers of the main constituent of the substance. The main constituent of the substance has the following structure:

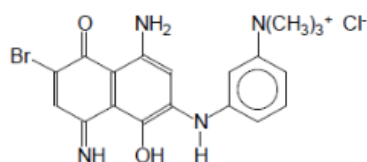


Figure 1. Main constituent of Basic Blue 99 constituents

3-[(4-amino-6-bromo-1-hydroxy-8-imino-5-oxo-5,8-dihydronaphthalen-2-yl)aminol-N,N,N-trimethylbenzenaminium chloride

The other major constituents of the substance are reported below:

Table 1. Structural formula and EC / IUPAC name of other major constituents of the substance

	Structural formula	EC / IUPAC name
1		3-((8-amino-7-bromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride
2		3-((8-amino-3,7-dibromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride
3		3-((8-amino-3,7-dibromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride
4		3-((8-amino-6,7-dibromo-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalen-2-yl)amino)-N,N,N-trimethylbenzenaminium chloride

Ref.: Submission V to SCCS- Basic Blue 99 -C059_- final

3.1.1.6 Empirical formula

New data

Table 2 Empirical formula of major constituents of the substance

	Structural formula	Empirical formula
Main constituent		C ₁₉ H ₂₀ BrN ₄ O ₂ ⁺ ·Cl ⁻ .
Other main constituents		

1.		$C_{19}H_{20}BrN_4O_2^+.Cl^-$
2.		$C_{19}H_{19}Br_2N_4O_2^+.Cl^-$
3.		$C_{19}H_{19}Br_2N_4O_2^+.Cl^-$
4.		$C_{19}H_{19}Br_2N_4O_2^+.Cl^-$

Ref: Submission V to SCCS- Basic Blue 99 -C059_ - final

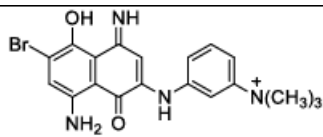
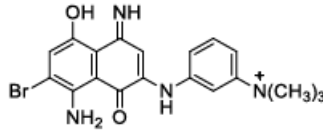
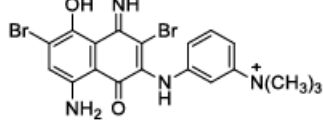
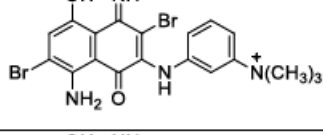
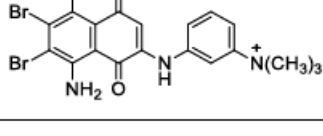
3.1.2 Physical form

Blue black, fine powder.

3.1.3 Molecular weight**New data**

Main constituents of the substance: MW = 451.8 g.mol⁻¹ (as chloride) and 416.3 g.mol⁻¹ (as cation)

Table 3. Molecular weight of other major constituents of the substance

1		451.8 g.mol ⁻¹ (as chloride) 416.3 g.mol ⁻¹ (as cation)
2		451.8 g.mol ⁻¹ (as chloride) 416.3 g.mol ⁻¹ (as cation)
3		530.7 g.mol ⁻¹ (as chloride) 495.2 g.mol ⁻¹ (as cation)
4		530.7 g.mol ⁻¹ (as chloride) 495.2 g.mol ⁻¹ (as cation)
5		530.7 g.mol ⁻¹ (as chloride) 495.2 g.mol ⁻¹ (as cation)

3.1.4 Purity, composition and substance codes

New data

The structural identity of the tested sample of 3-[(Bromo-8-amino-5-hydroxy-4-imino-1-oxo-1,4-dihydronaphthalenyl) amino]-N,N,N-trimethylbenzeneaminium chloride has been confirmed by NMR-spectroscopy for the recent batch VW080224A. The structural identity of the dye is additionally supported by the FTIR- and UV-spectra. Standard FTIR spectra of batch 74/75 and batch N° VW080224A were obtained over the wavenumber range 400-4000 cm⁻¹ and 500-4000 cm⁻¹ respectively, these FTIR spectra are comparable, and they are consistent with the proposed composition of the substance and with the chemical structure of the major constituents of the substance, i.e. 3-[(bromo and dibromo-4-amino--5,8-dihydro-1-hydroxy-8-imino-5-oxo-2-naphtyl)amino]-N,N,N -trimethylanilinium chloride.

According to the Applicant, the composition of the Basic Blue 99 has been determined by LC-UV (286 nm) and identification by LC/HRMS for the recent batches VW080224A, VW111124A and 5933503, as well as for the batch 74/75 (reference batch used in toxicity studies).

The results are described in Table 4 below:

Table 4. Content of the major constituents of the substance in batches VW080224A, VW111224A, 5933503 and 74/75.

Structure	VW080224A (area%)	VW111124A (area%)	9533503 (area%)	74/75
	40.52	37.62	40.73	52.90
	28.75	29.23	30.45	2.74
	6.35	5.51	4.12	6.08
	5.25	5.42	4.54	1.44
	4.32	5.00	4.41	1.80
TOTAL (area%)	85.19	82.78	84.25	64.96

According to the Applicant, based on these results (table 4), the knowledge on the composition of the substance as well as the quality of the substance have been significantly improved with the recent batches.

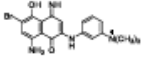
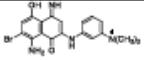
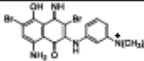
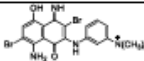
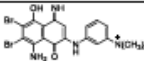
- The concentration of the main constituent of the substance Basic Blue 99 has been determined as 39.62% on average over the three recent batches.
- However, Basic Blue 99 is better characterized by the five (5) major constituents of the substance. The chemical structure of these major constituents has been fully elucidated by LCUV (276 nm), LC/HRMS and NMR.
- the purity of the substance is therefore determined based on the content of the five (5) major constituents which has increased, from of approximately 65% of the total composition of batch 74/75 to >80% of the total composition for batches VW080224A, VW111124A and 5933503. This is the result of improvements achieved in the new manufacturing process of the substance.
- The variability in the composition of the substance has been significantly reduced as demonstrated by the good reproducibility over the batches VW080224A, VW111124A and 5933503. This shows that the improvements made on the manufacturing process result in a better batch-to-batch consistency as previously reported.

Based on the results obtained on three batches, the purity of Basic Blue was determined as 84% on average over the three recent batches, which is significantly higher than the purity observed in the quality of the substance described in previous submissions.

The Applicant has also provided the following comparison with previous qualities of Basic blue 99:

In the past, the purity of the substance was already defined as the sum of several constituents. However, the peaks found in the HPLC profiles were not fully identified at the time and due to the different analytical methods used in the previous and the current submissions, the comparison of the new data with data obtained on previous batches in the past is difficult. Nevertheless, data on mono and dibrominated major constituents identified by previous analytical methods (without distinction between isomers) in different past batches have been compiled and reported in the table below for comparison.

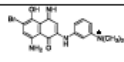
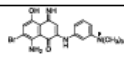
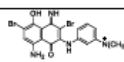
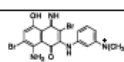
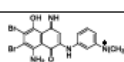
Table 5. Compilation of results obtained on previous batches (produced with **different manufacturing process**) using **different analytical methods**.

Chemical structure	Molecular formula and Molecular weight (λ_{max}) (as cation)	Previous analyses ** [%area]						
		** SCCS OPINION ON Basic Blue 99 (C059), SCCS/1537/14 (23/09/2014) and SCCS/1585/17(7 March 2017						
		batch 74/75 (2014) (2017)	Batch RS27988 101 (2014)	125 (2014)	140 (2014)	batch 106106 (2014) (2017)	Batch 107664 (2014)	batch 201501121 (2017)
	$C_{19}H_{20}BrN_4O_2^+$ 415 (625.58 nm)	62,8	50.2	48.2	57.3	64.1	67.8	61.3
	$C_{19}H_{20}BrN_4O_2^+$ 415 (625.58 nm)							
	$C_{19}H_{19}Br_2N_4O_2^+$ 493 (623.57 nm)	11,8	14.6	4.4	5.5	7.4	10.8	6.1
	$C_{19}H_{19}Br_2N_4O_2^+$ 493 (623.57 nm)							
	$C_{19}H_{19}Br_2N_4O_2^+$ 493 (623.57 nm)							
Purity		74.6 %	64.8 %	52.6 %	62.8 %	71.5 %	78.6 %	67.4%

The Applicant also notes that in Submission I and II, the purity of batch 74/75 was provided as 49.32% and the purity of batch RS27988101 was provided as 36.65%. In submission II (2014), the purity (48-67.8%) was provided as the sum of three peaks (415/417) with purity of batch 74/75 = 62.8%, batch RS27988101 = 50.2%, batch 125 = 48.2%, batch 140 = 57.3%, batch 106106 = 64.1% and batch 107664 = 67.8%. In addition, an analysis of the batch 74/75 (used in toxicity studies) has been conducted in the same conditions as the new batches in order to allow for the comparison with the new quality of Basic Blue 99 (expressed by the recent batches VW080224A, VW111124A and 5933503). The information about those former batches is cited to show that the purity was improved and the variability has been reduced.

According to the Applicant, thanks to the new analytical data, a new definition of the substance is now possible. This comparison is presented in table 6 below.

Table 6: Comparison of results obtained in the **new quality** (batches VW080224A, VW111224A, 5933503) with previous versions of Basic Blue 99, represented by batch 74/75. Data obtained with **the same analytical method and instrument**.

Chemical structure	Molecular formula and Molecular weight (λ_{max}) (as cation)	HPLC purity (area%) at 276 nm					
		batch VW080224A	batch VW111224A	batch 5933503	Proposed range	Proposed limit	batch 74/75
	$C_{19}H_{20}BrN_4O_2^+$ 415 (625.58 nm)	40.52	37.62	40.73	35-55	>35	52.90
	$C_{19}H_{20}BrN_4O_2^+$ 415 (625.58 nm)	28.75	29.23	30.45	25-35	>25	2.74
	$C_{19}H_{19}Br_2N_4O_2^+$ 493 (623.57 nm)	6.35	5.51	4.12	3-8	<10	6.08
	$C_{19}H_{19}Br_2N_4O_2^+$ 493 (623.57 nm)	5.25	5.42	4.54	3-7	<10	1.44
	$C_{19}H_{19}Br_2N_4O_2^+$ 493 (623.57 nm)	4.32	5.00	4.41	3-6	<10	1.80
Purity		85.49 %	82.78	84.25	–	≥80	64.96

According to the Applicant, based on these results, the purity of the substance has been significantly improved and the understanding of the structural composition of the substance has been further increased and backed by scientific data:

- The main constituents of the substance have been clearly identified (structures confirmed by NMR);
- The variability in the composition of the substance between the three new batches has been significantly reduced and new ranges/limits of major constituents' concentration have been defined to cover the new quality of Basic Blue 99.
- The purity of the substance was determined based on the content of the five (5) major constituents and has increased, from 64.96% of the total composition of batch 74/75 to ≥80% (results obtained for the three new batches). This is the result of improvements made on the manufacturing process of the substance.
- The results obtained with different analytical methods between this and previous submissions are difficult to compare. However, batch 74/75 has been analyzed (as a reference for previous submissions) together with the new batches to assure maximum consistency and comparability of the analytical data (table 6).

The Applicant concluded that the new manufacturing process has decreased the variability of the substance and increased the concentration of its major constituents, which now represent >80% of the substance.

Ref: Sensient Cosmetic Technologies. Dossier of hair dye C059_ batch 74 75. 2025.

SCCS comment

The SCCS agrees that the variability in the composition of the test substance has been reduced in the batches prepared by the new manufacturing procedure, and that the content of the five main constituents has been stabilized to greater than 80% in the new batches.

The SCCS notes that as shown in Table 6, there is a variation (tenfold difference) in the composition of the second main constituent between the new batches and the old batches. In particular, the second constituent is ~30% in the new batches compared to ~3% in the old batches used for toxicology studies.

The much higher levels of some characterised main constituents compared to the batch 74/75 used in the toxicity studies, means that these studies could not completely represent the toxicity of the new batches of Basic Blue 99.

The SCCS notes that in table 6 the correct purity of the batch VW080224A, derived by the sum of the five main components is 85.19 instead of 85.49.

3.1.5 Impurities / accompanying contaminants

New data

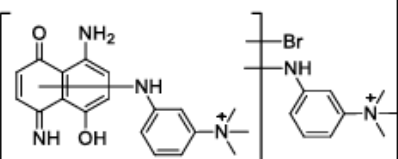
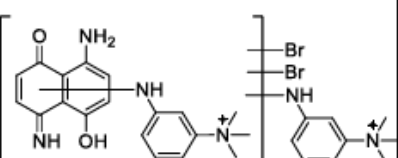
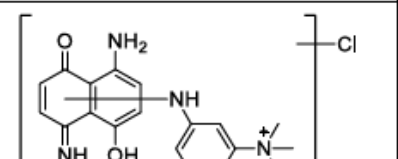
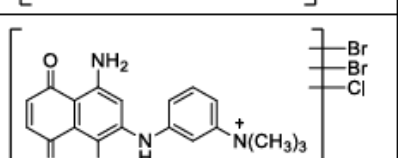
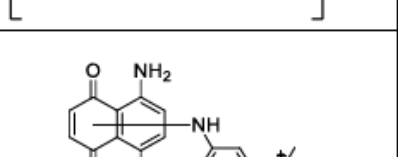
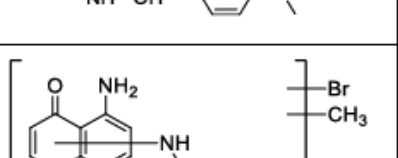
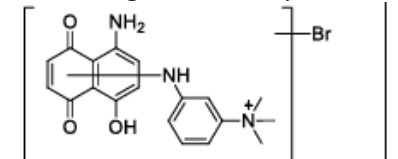
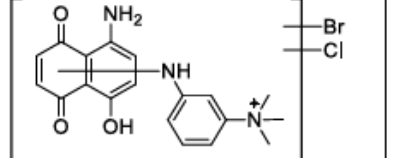
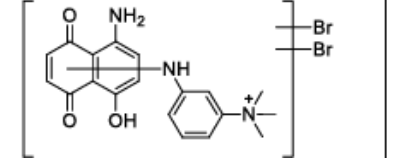
According to the Applicant, due to the changes made in the manufacturing process, a re-evaluation of the impurities present in the substance has been carried out (in parallel with the batch 74/75 used for the toxicological evaluation in the previous submission). Minor constituents, identified by LC/MS were reported as impurities in addition to the main constituents described in the previous sections. Finally, new ranges of minor constituents were reported to cover the new quality of Basic Blue 99. Compared to the batch 74/75, the total amount of impurities present in the substance was decreased from an average 35% (HPLC area%) to an average 15% in the recent batches. On average, only around 2% of the impurities remain to date unidentified.

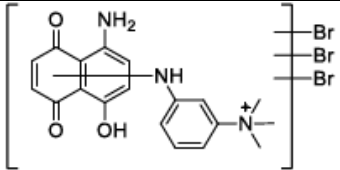
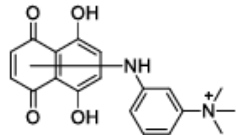
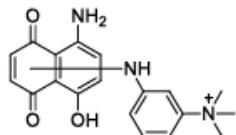
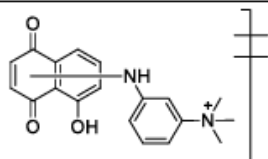
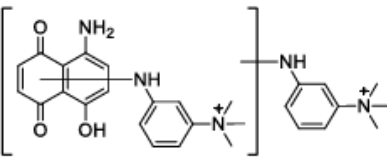
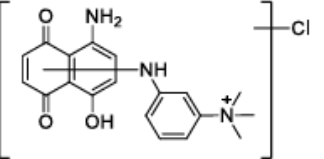
Table 7. Identified and non-identified impurities in 4 batches				
Impurities	VW080224A	VW111124A	5933503	74/75
Identified	13.3	15.08	13.35	31.18
Non-identified	1.43	2.14	2.40	2.40

The table 8 below summarises the identity of the impurities detected in Hair Dye C059.

Table 8. Composition of the substance and comparison with previous data

Structure	VW080224 A (area%)	VW111124 A (area%)	9533503 (area%)	74/75	Proposed Limits** (area%)
Major ($\geq 1\%$ in recent batches) known constituents having no impact on the toxicity of the substance <i>Average sum of impurities $\geq 1.0\%$ in recent batches: 8.02%</i>					
	5.24	8.21	6.95	0.22	< 10.0
	1.14	1.49	1.04	0.66	< 2.0
Major (<1% in recent batches) known constituents having no impact on the toxicity of the substance <i>Average sum of innocuous impurities <1.0% in recent batches: 3.67%</i>					
	0.86	0.83	0.76	1.84	< 1.5
	0.62	0.78	0.80	1.73	<0.2

	1.60	0.19	0.23	0.08	< 2.0
	0.57	0.86	0.52	0.17	< 1.5
	0.43	0.32	0.52	5.30	< 1.0
	0.20	0.22	0.18	–	< 0.5
	0.04	0.10	0.17	2.53	< 0.3
	0.05	–	0.16	3.25	< 1.0
Sum:	4.37%	3.30%	3.34%	14.9%	
Minor known constituents which may have an impact on the toxicity of the substance					
Average sum of impurities with potential toxicological impact in recent batches: 2.17%					
	0.96	0.58	0.60	9.59	< 2.0
	0.54	0.37	0.38	0.60	< 1.0
	0.35	0.37	0.35	1.93	< 1.5

	0.38	0.12	0.06	–	< 0.5
	0.12	0.22	0.24	0.79	< 0.5
	0.13	0.15	0.21	0.79	< 0.5
	0.08	0.08	0.07	–	< 0.2
	–	0.15	–	0.44	< 0.2
	–	–	–	1.08	< 0.2
Sum:	2.56%	2.04%	1.91%	15.22%	
Minor unidentified constituents					
Average sum of unidentified substances (all batches): 2.11%					
	VW080224 A (area%)	VW111124 A (area%)	9533503 (area%)	74/75	Proposed Limits** (area%)
Unidentified	1.43	2.14	2.40	3.86	<3.6
** Based on recent batches according to new manufacturing process.					

According to the Applicant, based on these results, the knowledge on the composition of the substance has been improved. All investigated constituents showed the same molecular backbones and substituents given in Figure 2. The substance is now overall better characterized as in the previous submissions. The content of unknown minor constituents has decreased, i.e. from 3.86 % of the total composition of batch 74/75 to an average 1.99% of the total composition of batches VW080224A, VW111124A and 5933503.

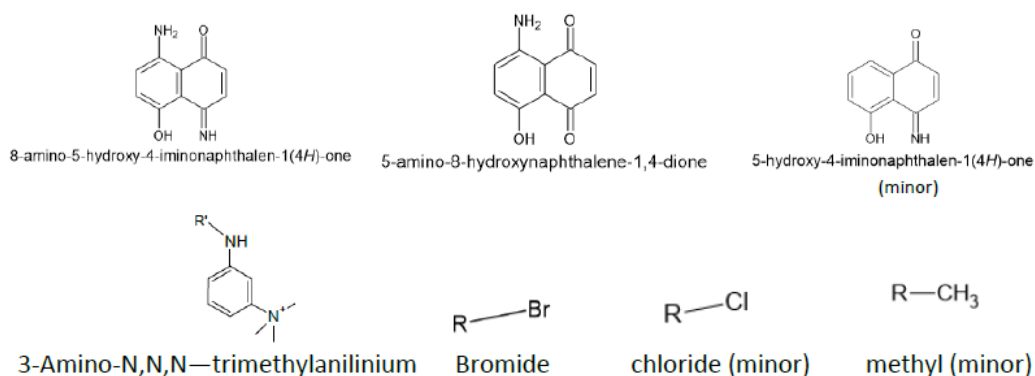


Figure 2. General structural moieties of Basic Blue 99 constituents

Following a request from the SCCS, the Applicant provided additional data related to the in-silico assessment of impurities' toxicity.

***In silico* predictions**

To characterize the genotoxic and carcinogenic potential of the newly identified impurities in Basic Blue 99, a targeted in silico strategy was implemented using four complementary tools, each contributing a distinct predictive dimension. Initially, all impurities were screened using the OECD QSAR v4.5 Toolbox and Toxtree v3.1.0.

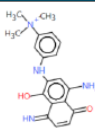
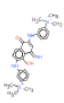
After screening, the impurities were grouped into families based on their structural similarity ($\geq 70\%$ similarity using OECD QSAR Toolbox clustering) and the presence of relevant functional groups such as aromatic amines, quinones, or other known toxicophore groups. Impurities within each family were further evaluated based on their toxicophore profile, ensuring that each group reflected compounds with similar functional characteristics that could potentially lead to similar toxicological effects. For each representative impurity selected from the families, DEREK Nexus was used to identify potential genotoxicity alerts. In cases where alerts were identified, Leadscope, a statistical QSAR model, was employed as a cross-validation tool.

Impurities were grouped into major and minor families based on their molecular backbone, functional characteristics, and concentration. The in-silico prediction results for representative molecules from each family are presented in the following table.

- **Major Impurities ($\geq 1\%$ and $< 1\%$ in recent batches) with No Significant Toxicity**

The backbone of the impurities in the major families is represented by Impurity 3, as shown in the following table. This structure contains an aromatic amine group and α, β -unsaturated carbonyls. However, these features do not present significant genotoxicity or carcinogenicity concerns in this case, as confirmed by the in silico analysis.

Table 9 Example of In Silico Prediction Results for Representative Impurities in Basic Blue 99

Substance	Structure	SMILES	OECD QSAR Toolbox	Nexus DEREK	Leadscope	Consensus
Impurity 3		<chem>C1=CC=C(C=C1NC(=O)C(=O)C(=O)N)N</chem>	Toxicity Hazard Classification by Cramer: High (Class III) Alerts found - Carcinogenicity (genotox and nongenotox): Primary aromatic amine, hydroxyl amine and its derived esters (Genotox); Structural alert for genotoxic carcinogenicity. - <i>In vitro</i> mutagenicity (Ames test): Primary aromatic amine, hydroxyl amine and its derived esters. - <i>In vivo</i> mutagenicity (Micronucleus): H-acceptor-path3-H-acceptor; Primary aromatic amine, hydroxyl amine and its derived esters.	- Mutagenicity in Bacteria: negative - Mutagenicity in vitro: no alerts found - Chromosomal aberration in vitro: no alerts found. - Genotoxicity in vivo: no alerts found - Carcinogenicity: PLAUSIBLE (Alert found: alpha, beta-unsaturated aldehyde, ketone or imine)		Genotoxicity: negative Carcinogenicity: alerts not relevant for human.
Impurity 2		<chem>C[N+](C)(C)C1CCC(NC2=CC(=O)C(=O)C2)C1</chem>	Toxicity Hazard Classification by Cramer: High (Class III) - Carcinogenicity (genotox and nongenotox): no alerts - <i>In vitro</i> mutagenicity (Ames test): no alerts - <i>In vivo</i> mutagenicity (Micronucleus): H-acceptor-path3-H-acceptor; Primary aromatic amine, hydroxyl amine and its derived esters.	- Mutagenicity in Bacteria: Plausible - Mutagenicity in vitro: no alerts found - Chromosomal aberration in vitro: no alerts found. - Genotoxicity in vivo: no alerts found - Carcinogenicity: No alerts	Mutagenicity in bacteria: negative	Genotoxicity: negative Carcinogenicity: negative

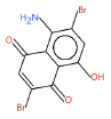
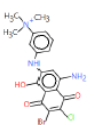
The major families of impurities, whether present at $\geq 1\%$ or $< 1\%$, share the same molecular backbone and do not present significant genotoxic or carcinogenic concerns for humans. For mutagenicity, Impurity 3 showed no genotoxicity alerts in mutagenicity in bacteria tests, with negative results confirmed by DEREK Nexus. On the other hand, Impurity 2 showed positive mutagenicity results in bacteria, which were confirmed by Leadscope, but DEREK Nexus predicted negative genotoxicity results. Regarding carcinogenicity, both impurities triggered plausible alerts due to the presence of α,β -unsaturated aldehydes and ketones in their structures. However, these alerts are not relevant for humans as they are based on rodent studies showing effects on the forestomach, which is not present in humans.

According to the Applicant, the major families do not raise significant concerns for genotoxicity or carcinogenicity in humans, with Impurity 2 showing positive mutagenicity in bacteria but no carcinogenic effects relevant to humans.

• Minor Families (with Quinone Group)

The minor families contain impurities that include quinone groups, which are associated with genotoxicity alerts. These impurities are present at lower concentrations ($< 1\%$) but raise concerns regarding genotoxicity and carcinogenicity. The two examples from this family, shown in the table, are below:

Table 10: In Silico Prediction Results for Example Molecules in Minor Families with Quinone Groups

Substance	Structure	SMILES	OECD QSAR Toolbox	Nexus DEREK	Leadscope	Consensus
Impurity 6		<chem>Nc1c(Br)cc(O)c2C(=C(Br)=C(C(=O))c12)=O</chem>	Toxicity Hazard Classification by Cramer: High (Class III) - Carcinogenicity (genotox and nongenotox): Quinones (Genotox) ; Structural alert for genotoxic carcinogenicity - <i>In vitro</i> mutagenicity (Ames test): alerts found (quinones) - <i>In vivo</i> mutagenicity (Micronucleus): H-acceptor-path3-H-acceptor .	Carcinogenicity: plausible (Alerts: alpha, beta-unsaturated aldehyde, ketone or imine) Mutagenicity in bacteria: plausible (quinones) Chromosome damage <i>in vitro</i> : plausible (quinones) Mammalian cells mutagenicity: no alerts Mutagenicity <i>in vivo</i> : no alerts Chromosome damage <i>in vivo</i> : equivocal	Carcinogenicity: negative Mutagenicity in Bacteria: positive Chromosomal aberration <i>in vitro</i> : positive Chromosomal aberration <i>in vivo</i> : positive Mammalian cells mutagenicity: positive Mammalian mutagenicity <i>in vivo</i> : Not in domain	Carcinogenicity: negative Genotoxicity: alerts (Quinones)
Impurity 2		<chem>C[N+](C)(C)c1cc(ccc1)Nc1cc(N)c2C(=O)C(Cl)=C(Br)C(=O)c2c1O</chem>	Toxicity Hazard Classification by Cramer: High (Class III) - Carcinogenicity (genotox and nongenotox): Quinones (Genotox) ; - <i>In vitro</i> mutagenicity (Ames test): alerts found (quinones) - <i>In vivo</i> mutagenicity (Micronucleus): H-acceptor-path3-H-acceptor .	Carcinogenicity: plausible (Alerts: alpha, beta-unsaturated aldehyde, ketone or imine) Mutagenicity in bacteria: plausible (quinones) Chromosome damage <i>in vitro</i> : plausible (quinones) Mammalian cells mutagenicity: no alerts Mutagenicity <i>in vivo</i> : no alerts Chromosome damage <i>in vivo</i> : equivocal		Carcinogenicity: negative Genotoxicity: alerts (Quinones)

The minor families of impurities, which contain quinone groups, show genotoxicity alerts. These alerts are associated with the quinone group, which is known for its genotoxic potential in certain *in silico* and *in vitro* tests. However, the quinone group is not classified as genotoxic under Annex VI of the CLP regulations. Regarding carcinogenicity, the alerts for both Impurity 6 and Impurity 2 were classified as plausible due to the presence of α,β -unsaturated aldehydes and ketones. However, these alerts are not relevant for humans, as discussed earlier, because of the forestomach effects observed in rodent studies, which are not applicable to human risk assessment. According to the Applicant, while the quinone-containing impurities show genotoxicity alerts, the quinone group itself (CAS: 106-51-4) is not classified as genotoxic under Annex VI of the CLP, and the carcinogenicity alerts are considered irrelevant for humans.

Applicant's general conclusion and impact on the classification of Basic Blue 99:

The *in silico* evaluation of Basic Blue 99 impurities shows that the major families do not present significant genotoxicity or carcinogenicity concerns for humans. While Impurity 2 showed positive mutagenicity in bacteria, these alerts, along with similar alerts in other impurities, were reviewed by experts and alternative models (such as Leadscope), and were ultimately considered not relevant for human health. For the minor families containing quinone groups, genotoxicity alerts were observed but, as the quinone (CAS: 106-51-4) is not classified as genotoxic under Annex VI of the CLP, and the carcinogenicity alerts were determined to be irrelevant for humans, given the rodent-specific nature of the findings. As a result, Basic Blue 99 does not require changes to its current classification, as the impurities evaluated do not pose significant risks for human health.

Ref: Sensient *In silico* evaluation of impurities in Basic Blue 99

According to the Applicant, most impurities raise no toxicological concern. The only structural element of potential concern is the quinone moiety, but the content of minor known constituents which may have an impact on the toxicity of the substance has decreased, i.e. from 15.22% of the total composition of batch 74/75 to an average of 2.17% of the total composition of batches VW080224A, VW111124A and 5933503. In particular, 3-[(bromo-4-amino-5,8-dihydro-1-hydroxy-5,8-dioxo-2-naphthyl)amino]-N,N,N-trimethyl-anilinium chloride was present at 9.59% in batch 74/75 while it is present at <1.0% in all three new test batches. Therefore, it is considered as a minor constituent (impurity) in the new quality while it was a major constituent in the previous batches. A specific attention has been paid to the impurities of the new manufacturing process, and specifically to the quinone-containing

structures, which are well-known for their toxicological concern. The decrease of this concentration is consistent with a better quality of BB99. The substance used in physical and chemical tests, and particularly in toxicity studies, mentioned in the current opinion is representative of the substances present in commercial products. Some of the batches used in toxicological studies can be considered as worst cases, since they contain overall higher concentrations of impurities and especially impurities of potential toxicological concern. According to the Applicant, the assays performed on Batch RS27988101 can also be used to support the safety of Basic Blue 99 in a weight of evidence approach, according to the general rules of toxicology and to the latest SCCS's Notes of Guidance. The batches RS27988101 (purity 64.8%) and 74/75 (purity 74.6%) have been used in some of the toxicity studies; they contain more minor constituents, including those containing the quinone moiety, than the more recent batch VW080224A. Therefore, those batches are considered as a 'worst case', especially regarding the genotoxicity studies. However, negative results in relevant tests (human cells, *in vivo*) have been observed with both batches. Decreasing the concentration and the number of impurities improves the quality of the substance and confirms the absence of toxicological concern. The summary table (Table 11) reports the batches used in the different toxicological studies.

Table 11. List and Results of toxicology studies

Test	Date	Laboratory	Sponsor	OECD/ Guideline	GLP Compliance	Batch	Concentration Tested	Results
Acute toxicity	Feb 1986	Huntingdon RC 86131 D		OECD 401		KS3993		LD 50 > 2000 mg/kg (rats; p.o)
Acute Eye irritation	Aug 2004	RCC Ltd Switz / 853987	Henkel KGaA	OECD 405 April 2002	YES	RS27988101	Undiluted	Severely irritating (Rabbit eye, undiluted)
Acute Skin irritation	Aug 2004	RCC Ltd Switz / 853986	Henkel KGaA	OECD 404 April 202	YES	RS27988101	Undiluted	not irritating (Rabbit Skin, undiluted)
Skin sensitization (LLNA)	Feb 2001	RCC Ltd Switz / 795363	Henkel KGaA	OECD 406 July 1992	YES	RS27988101	1%; 5%; 25%	non sensitizer (mouse, up to 25 %)
In vitro Mutagenicity - Ames test	June 2005	GenPharm Tox / 100128	Henkel KGaA	OECD TG 471	YES	74/75	0.003.16-1.0 mg/plate w/wo S9	Positive (+/+) w/wo S9 Possible Dose effect
In vitro Mutagenicity - Ames test	April 2024	Genevolution / 3714-316	Sensient	OECD TG 471	YES	VW080224A	20-5000 µg/plate w/wo S9	Positive (+/+) w/wo S9 Possible Dose effect
In vitro Mutagenicity - hprt test	June 2024	Labcorp / 8530267	Sensient	OECD TG 476	YES	VW080224A	0.75-30 µg/ml w/wo S9	Negative
In vitro Mutagenicity, Mouse Lymphoma Assay (TK+/-)	Dec 2005	RCC CCR GmbH 897002	Henkel KGaA	OECD TG 476 1997	YES	74/75	wo S9 1,3- 20µg/ml w S9 5,0- 40µg/ml	Negative (-/-)
In vitro Mutagenicity, Micronucleus test	Sept.05	GenPharm Tox / 100135	Henkel KGaA	OECD TG 487 2004	YES	74/75	wo 1.56- 6.25µg/ml 37.5-150µg/ml	Positive (+/+) w/wo S9 Dose effect with S9

In vivo Mutagenicity, Micronucleus test	Dec 2005	RCC - CCR GmbH / 909502	Henkel KGaA	OECD 474 July 1997	YES	74/75	1.25-5.0mg/kg	Negative on MNT mouse on Bone Marrow PCE
In vivo/in vitro Unscheduled DNA Synthesis	Sept.03	RCC CCR GmbH 792600	KPSS GmbH	OECD 486 EC directive 2000/32 B39	YES	RS27988101	2 & 16 hours at 1000 & 2000 mg/kg bw.	Negative
Subchronic oral toxicity 13 week 4 week recovery	January 2006	RCC Ltd Switz A16470	Henkel KGaA	OECD 408 / Sept 1998 EC Directive 96/54 Sept	YES	74/75	0,10,60, 240mg/kg bw./d	NOAEL 60 mg/kg.bw/d <u>Recalculated NOAEL: 37 mg/kg.bw/d with purity 62.8</u>
Developmental/ Reproductive toxicity	January2006	RCC Ltd Switz A16481	Henkel KGaA	OECD 414 April 2004 EEC Guideline 2004/73 B31 April2004	YES	74/75	0,60,200,500mg /kg bw/day	NOAEL (embryofoetal) at 200mg/kg.bw/d NOEL At 60mg/kg.bw/d <u>Recalculated NOAEL: 37 mg/kg.bw/d with purity 62.8</u> No teratological or embryoethal effect at 500mg/kg.bw/day
Skin penetration (in vitro)	Dec 2002	CTL JV1718	Henkel KGaA	In vitro Dermatomed Pig Skin	YES	RS279881 01		approx. 0,8 µg/cm2

Ref: CEHTRA (2023); Internal data file (2023); Lhasa Limited (2022); Leadscope Inc. (2022); Ideaconsult Ltd. (2020); Lhasa Limited (2023); Leadscope Inc. (2023); NTP (2003).

SCCS comments

The SCCS agrees that the total amount of impurities was decreased from ~35% in the old batches to ~15% in the new batches prepared by the new manufacturing procedure. However, there is variation in the impurity profiles between the new batches and Batch 74/75, which was used in previous toxicity studies. Some impurities categorised by the Applicant as impurities with no impact on the toxicity of the test substance, have significantly increased in the new batches, while the minor impurities categorised by the Applicant with impact in the toxicity have been decreased. The average sum of impurities with potential toxicological impact in recent batches is around 2.17% (proposed limit by the Applicant < 3.6 %, is much higher than the actual values), and the average sum of unidentified impurities is 1.99 % in the new batches. The Applicant has therefore claimed that the reduction of certain impurities of toxicological concern, together with the absence of concern for the remaining ones, mitigates any toxicological risk. However, this statement is not supported by the *in silico* studies submitted, which lack robustness and fail to substantiate the Applicant's conclusions. In Table 7, the content in terms of % area w/w, for the non-identified impurities in the batch 74/75, is 3.86 %, and in Table 8 the average sum of unidentified impurities for all batches is 2.45 %.

The discussion on the *in silico* assessment of impurities lack some important points and it is not clearly presented. References to individual impurities are unclear, and for instance impurity 1 (quinone) is not clearly mentioned. The reports from the OECD QSAR Toolbox are missing. The Applicant has also referred to "Impurity 1, Impurity 2, etc." without providing a clear link to the corresponding chemical structures already submitted. The results from the DEREK and Leadscope models are only summarized without discussing the information on the applicability domain and disregard some results present in the reports. For instance, deeper discussion is necessary in these cases: for Impurity 1 DEREK results indicate that carcinogenicity in humans is plausible. Several alerts for different toxic effects are indicated by DEREK. For Impurity 2 results from DEREK indicate plausible genotoxicity and carcinogenicity. From the Leadscope models there are substances similar with values >40% which are mutagenic. For Impurity 3 results from Derek indicate plausible carcinogenicity and skin sensitisation. For Impurity 6 results from Derek indicate plausible genotoxicity and carcinogenicity. Applicant states that "Leadscope is used to confirm or refine the alerts observed in DEREK Nexus and OECD QSAR Toolbox." This perspective is not logical, since Applicant used the statistical tools in Leadscope, and not the models based on structural alerts. It is unclear why certain models were selected from DEREK and others from Leadscope,

since both systems offer statistical and rule-based models. Furthermore, the strategy to confirm results using a second orthogonal system is not correct. Indeed, the consolidated approach (see the ICH M7 strategy) is to use both statistical and expert-based models and is to assume a positive effect if at least one of the two approaches gives a positive result. The fact that in some cases the models are indicating that the information on possible effects is insufficient does not mean that there is lack of toxicity.

Other characteristics of hair dye C059:

The test substance was further characterized by several analytical methods. Trace heavy metal contents have been determined by ICP-OES. Weight-% fractions of anions such as chloride, bromide, sulfate, methylsulfate and acetate were determined by ion-chromatography. Insoluble and volatile matter residuals have been determined gravimetrically. The water content was determined by Karl-Fischer-Titration. The residual ethylene glycol content in the sample of batch VW080224A was determined by GC/MS. The volume specific surface area (VSSA) was determined for the batch 5933503.

Table 12. Supplementary analytical data supporting the characterization of hair dye C059.

Type of Analysis	Measured variable	Results				
		VW080224A	VW111124A	5933503	74/75 ^{a)}	Proposed range
Ion chromatography	Chloride (%)	23.1	19.3	17.2	20.2	< 25.0
	Bromide (%)	n.a.	n.a.	n.a.	n.a.	not present
	Sulphate (%)	0.08	trace	trace	0.53	<0.2
Karl-Fischer	Water content (%)	4.94	3.77	5.48	7.85	<8.0
Gravimetry	Insoluble matter (%)	0.71	0.38	2.55	0.45	<5.0
Thermogravimetry	Volatile matter	2.17	2.50	4.11	n.a.	<8.0
GC/MS	Ethylene glycol content (ppm)	7200	71	602	n.a.	<620 ^{b)}
ICP/OES	Antimony (ppm)	0.0	0.0	0.0	<1.0	<10
	Arsenic (ppm)	0.0	0.0	0.0	<1.0	<3
	Barium (ppm)	0.0	0.0	0.0	1.0	<20
	Cadmium (ppm)	0.0	0.0	0.0	<1.0	<1
	Chromium (ppm)	1.0	0.0	0.0	39	<20
	Cobalt (ppm)	0.0	0.0	0.0	<1.0	<20
	Copper (ppm)	5.0	0.0	5.0	18.0	<20
	Iron (ppm)	14.0	41.0	207.0	6900	<250
	Lead (ppm)	2.0	0.0	0.0	1.0	<10
	Manganese (ppm)	0.0	2.0	3.0	n.a.	<20
	Mercury (ppm)	0.0	0.0	0.0	1.0	<1
	Nickel (ppm)	1.0	0.0	0.0	5.0	<10
	Selenium (ppm)	0.0	0.0	0.0	n.a.	<10
	Zinc (ppm)	33100	49000	55000	66250	<70000
VSSA	VSSA (m ² /cm ³)	n/d	n/d	2.79	n.a.	<6

^{a)} Data reported from dossier of hair C059 (submission III, 2013). ^{b)} Limit recommended by ICH guideline Q3C (R8) on impurities: guideline for residual solvents, version 29/04/2024.

Ref: Dossier of hair dye C059_batch VW080224A_VW111124A_5933503: Study Reports 56059-1, 56059-2, 56059-4, 56059-5, 56059-6, 56059-7, 56059-8

3.1.6 Solubility

New data

Water 10-100 g/L room temperature
Ethanol 1-10 g/L room temperature
DMSO 1-10 g/L room temperature

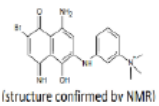
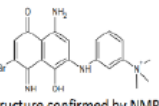
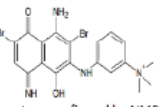
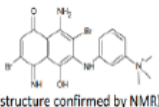
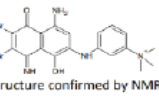
Ref. : Sensient Cosmetic Technologies. Dossier of hair dye C059_ batches VW080224A, VW111124A, 5933503, 2025

3.1.7 Partition coefficient (Log P_{ow})

New data

The LogP_{ow} are predicted for the following structures.

Table 13: Partition coefficient of the major constituents of the substance

Chemical structure	Molecular composition and monoisotopic mass of structure suggested	Molecular formula	Log Kow (KOWWIN v1.68 estimate)
 (structure confirmed by NMR)	C ₁₉ H ₂₀ BrN ₄ O ₂ 415,076407	C ₁₉ H ₂₀ BrN ₄ O ₂	1,12
 (structure confirmed by NMR)	C ₁₉ H ₂₀ BrN ₄ O ₂ 415,076407	C ₁₉ H ₂₀ BrN ₄ O ₂	1,12
 (structure confirmed by NMR)	C ₁₉ H ₁₉ Br ₂ N ₄ O ₂ 492,986912	C ₁₉ H ₁₉ Br ₂ N ₄ O ₂	2,01
 (structure confirmed by NMR)	C ₁₉ H ₁₉ Br ₂ N ₄ O ₂ 492,986912	C ₁₉ H ₁₉ Br ₂ N ₄ O ₂	2,01
 (structure confirmed by NMR)	C ₁₉ H ₁₉ Br ₂ N ₄ O ₂ 492,986912	C ₁₉ H ₁₉ Br ₂ N ₄ O ₂	1,51

Ref: Submission V to SCCS- Basic Blue 99 -C059_ - final

3.1.8 Additional physicochemical specifications

New data

Melting point:	> 200 °C (Ref 2)
Boiling point:	No data available
Flash point:	Not applicable
Vapour pressure:	No data available
Density:	1.71 g/cm ³
Specific surface area (SSA)	1.63 m ² /g
Viscosity:	Not applicable
pKa:	No data available
Refractive index:	No data available
Optical rotation:	Not relevant, the substance does not contain identified constituents with chiral center.

UV-Vis spectrum (200-800 nm): UV-Vis spectra of batch 74/75 and batch N° VW080224A were obtained over the wavenumber range 200-700 cm⁻¹ after dissolution of the test item in deionized water pH 7 and measured in a concentration of 25 mg/1000mL. The measuring UV-VIS spectra of batch 74/75 and VW080224A are comparable and show maximum absorption at the wavelength of 268/269 nm, 578/576 nm and 620/616 nm respectively.

3.1.9 Homogeneity and Stability

From Previous opinion (SCCS/1585/17)

A freshly prepared sample of Basic Blue 99 batch 0107664 at 0.05 mg/ml in water was compared by HPLC-DAD with a sample stored 3 days at autosampler conditions (4°C). According to the main peak area, the sample was stable within a period of 3 days at 4°C, as a recovery of 99.6 % was found under the study conditions.

New data

The sample solutions of batches 5933503, VW111124A and VW080224A was re-analyzed after storage at 20 °C for 7 days to assess its stability with respect to the main constituents. After 7 days at 20 °C, no degradation was observed, therefore, a good stability of the substance is expected. According to the Applicant, the percentage of recovery indicates a good stability in studied conditions, aligned with previous submissions.

SCCS comment

The study report, both methodology and detailed data, of the stability results reported by the applicant was not submitted.

The SCCS notes that stability studies of the test substance under the conditions of use must be performed by using the same HPLC-PDA method (at λ_{max}) used to analyse impurities.

Overall SCCS comments in the Physicochemical Part

1. The SCCS agrees that the variability in the composition of the test substance has been reduced in the batches prepared by the new manufacturing procedure, and that the content of the five main constituents has been stabilized to greater than 80% in the new batches.
2. The SCCS notes that there is a variation (tenfold difference) in the proportion of the second main constituent between the new batches and the old batches. In particular, the second main constituent is ~30% in the new batches compared to ~3% in the old

batches used for toxicology studies. There is also variation in the impurity profiles between the new batches and Batch 74/75, which was used in previous toxicity studies. Some impurities have significantly increased in the new batches. Thus, the toxicological data that are based on Batch 74/75 or similar old batches (RS27988101), does not fully correspond to the material specifications outlined in the current Submission.

3. The *in silico* assessment of the impurities lacks clarity, key information, and a scientifically sound methodology. The SCCS considers that the Applicant's claim of reduced toxicological concern for impurities is not substantiated by the *in silico* studies submitted. Based on the submitted data, for the new batches, the average sum of impurities with potential toxicological impact is 2.17% and the average sum of unidentified impurities is 1.99% in the new batches. In total, this corresponds to approximately 4.16% of impurities for which a TTC approach would be applicable.
4. The SCCS notes that stability studies of the test substance under the conditions of use must be performed by using the same HPLC-PDA method (at $\lambda_{\max}$) used to analyse impurities and full reports should be submitted.

3.2 Function and uses

From Previous opinion (SCCS/1585/17)

Basic Blue 99 is intended to be used as a direct hair dye (semi-permanent hair dye) in hair colouring products with a maximum on the scalp up to a concentration 1.0%. A monthly use is expected, for about 30 minutes.

3.3 Toxicological evaluation

3.3.1. Acute toxicity

3.3.1.1. Acute oral toxicity

Guideline:	OECD 401
Species/strain:	CD Rat Sprague-Dawley
Group size:	5 male and 5 female rats
Test substance:	Basic Blue 99 prepared in 20% w/v in distilled water
Batch:	KS 3993
Purity:	not reported
Vehicle:	distilled water
Dose levels:	Single dose 2 g/kg Body weight in volume of 10 ml/kg
Observation period:	15 days
Administration:	syringe and plastic catheter
GLP:	Not in compliance
Study period:	February 1986

One Group of 5 male and 5 female rats received a single oral dose of 2 g/kg bw. No control animal was included in the study. The animals were observed twice a day for 15 days for mortality and clinical abnormalities. Body weights and macroscopic observation were recorded, but histological examinations were not performed.

Results

Reaction to treatment observed shortly after dosing were piloerection, abnormal body carriage (hunched posture), abnormal gait (waddling) and increased salivation. Bodyweight gains were recorded for all rats on Day 8 and 15. Total recovery measured by external

appearance and behavior was complete after 3 days. No abnormalities were recorded at autopsy.

Conclusion

The acute lethal oral dose of Basic Blue 99 was found greater than 2 g/kg.bw.

This batch has not been characterized, but the information is cited as supportive information.

Ref: Kynoch SR. Acute Oral Toxicity to Rats of Arianor Steel Blue, Report No. R 9501207, 1986

SCCS comment

The test substance, batch KS 3993, is not representative with the material specifications outlined in the current Submission

3.3.1.2. Acute dermal toxicity

/

3.3.1.3. Acute inhalation toxicity

/

3.3.1.4. Acute intraperitoneal toxicity

/

3.3.2. Irritation and corrosivity

3.3.2.1. Skin irritation

Guideline:	OECD 404 (2002)
Species/strain:	New Zealand white rabbit, SPF
Group size:	3 animals (1 male and 2 female)
Test substance:	C059
Batch:	RS27988101
Purity:	Not reported in the study report but 50.2% (HPLC, based on area – Dossier submitted to SCCS on April 2013)
Vehicle:	purified water
Dose level:	0.5 g of C059 Undiluted (a black solid item); pH of 1% solution: 3.10
Dose volume:	
Observation:	
GLP:	In compliance
Study period:	June 2004

The substance was applied by topical semi occlusive application of 0.5 g to the intact left flank of each of the three young adult New Zealand White rabbits. The duration of the treatment was four hours. The scoring of the skin reaction was performed 1, 24, 48, and 72 hours as well as 7, 10, and 14 days after removal of the dressing. The mean score was calculated across 3 scoring time (24, 48 and 72 hours after patch removal) for each animal for erythema / eschar grades and for oedema grades separately.

Results

The mean erythema / eschar score and the mean oedema score was 0 for all three animals. The application of the C059 to the skin resulted in no signs of irritation. However slight blue staining was observed in all animals from one hour reading to 14 days after treatment, the

end of the observation period of all animals. No corrosive effects were noted on the treated skin of any of the measuring intervals and no clinical signs were observed.

Thus the test item, did not induce significant or irreversible effect damage to the skin.

Conclusion

Under the condition of the test the study authors considered that C059 to be not irritating to rabbit skin.

Ref: Arcelin G. C 059: Primary Skin Irritation Study in Rabbits, Report No. R 0400811, 2004.

SCCS comment

The test substance, batch C059, is not representative of the material specifications outlined in the current Submission.

3.3.2.2. Mucous membrane irritation / Eye irritation

Guideline:	OECD 405 (2002)
Species:	New Zealand white rabbit, SPF
Group:	1 animal (1 female) (suspected that C059 might produce irritancy)
Test Substance:	C059
Batch:	RS27988101
Purity:	Not reported in the study report but 50.2% (based on area – Dossier submitted to SCCS on April 2013)
Dose:	0.1 g of C059 Undiluted (a black solid item) pH of 1% solution: 3.10
GLP:	In compliance
Study period:	June 2004

The substance was placed in the conjunctival sac of the left eye of a single young adult New Zealand White rabbit. The right eye remained untreated and served as the reference control. The treated and untreated eyes were rinsed with lukewarm tap water after 24 hours. The mean score was calculated across 3 scoring time 24, 48 and 72 hours after instillation for the animal for corneal opacity, iris, redness and chemosis of the conjunctiva, separately.

Results

The individual mean score for corneal opacity was 3.00 and for the iris the maximum score of 2.00. The reddening of the conjunctivae was not assessable during all observation period due to the staining of the item and individual mean score for chemosis was 3.67. The instillation of C 059 into the eye of one animal resulted in severe and early-onset ocular changes, discharge and chemosis. The assessment of the redness of the conjunctivae and sclera was not possible throughout the observation period due to staining produced by the test item. The assessment of the Cornea and iris was also not possible 1 and 24 hours after treatment due to staining produced by the test item. Marked opacity, affecting the whole area, was noted in the eye at the 48- and 72-hour reading as well as an absent light reflex.

Marked black to blue staining of the treated eye by the test item was evident from the 1 – hour reading up to 72 hours after treatment. No clinical signs were observed. The Sponsor and Study Director agreed to close the study with one animal treated after 72 hour reading

Conclusion

Possible corrosion cannot be concluded (Study prematurely closed due to the reported result).

Ref: Arcelin G. C 059: Primary Eye Irritation Study in Rabbits, Study Number 853987_8

SCCS comment

Although the test was performed in one animal, it points towards an eye-irritation potential of the test substance. An eye irritant effect due to physical shape (black solid) of the test item cannot be ruled out.

The test substance, batch C059, is not representative of the material specifications outlined in the current Submission.

3.3.3. Skin sensitisation

Local Lymph Node Assay (LLNA)

Guideline: OECD 406 (1992)
 Species: Mice CBA / JNCRJBR
 Group: 3 test and 1 control group; 4 females per group
 Substance: ARIANOR STEEL BLUE
 Batch: RS27988101
 Purity: superior to 55 FL% (HPLC)
 Dose: test substance: 1, 5 and 25% (w/v) in bi-distilled water,
 Vehicle: test substance: Bi-distilled water
 Control: Bi-distilled Water
 GLP: in compliance
 Study period: Feb 2001

Three groups of four female mice were treated daily with the test item at concentrations of 1, 10 and 25% in bi-distilled water by topical application to the dorsum of each ear lobe (left and right) for three consecutive days. The concentration of 25% was selected as the highest technically applicable concentration in the chosen vehicle. A control group of four mice was treated with the vehicle (bi-distilled water) only,

Five days after the first topical application, the mice were injected intravenously into a tail vein with radio-labelled thymidine (^3H -methyl thymidine).

Approximately five hours after intravenous injection, the mice were sacrificed, the draining auricular lymph nodes excised and pooled per group. Single cell suspensions of lymph node cells were prepared from pooled lymph nodes which were subsequently washed and incubated with trichloroacetic acid overnight. The proliferative capacity of pooled lymph node cells was determined by the incorporation of ^3H -methyl thymidine measured in a β scintillation counter

Results

All treated animals survived the scheduled study period. No clinical signs were observed. The results obtained (Stimulation index (S.I.)) are reported in the following table. The estimated concentration of test item required to produce a S.I. of 3 is referred as the EC3 value.

Table 14 Results of Local Lymph Node Assay

	Test item concentration %	S.I.
Group 2	1	0.7
Group 3	5	1.1
Group 4	25	1.1

Based on these criteria, ARIANOR STEEL BLUE is regarded as a non-sensitizer.

Conclusion

The study authors considered Arianor Steel Blue was found to be a non-sensitizer.

SCCS comments

According to the OECD guideline 429 (adopted July 2010), a wholly aqueous test substance should be avoided. The test substance, batch RS27988101, is not representative with the material specifications outlined in the current Submission.

There are several reports in the public literature documenting cases of delayed-type and immediate-type contact allergic reactions in humans. See 3.3.8 Human data.

The SCCS considers Basic Blue 99 to be a skin sensitiser.

3.3.4. Dermal / percutaneous absorption

Guideline:	OECD 428 (2000) - Based on the guidance notes of the EU SCCNFP (2000) OECD Draft Guidelines on Percutaneous absorption (2000) and ECETOC recommendation (1993)
Species/strain:	dermatotomed pig skin, 400 µm thickness
Membrane integrity:	6 glass diffusion cells (2.54 cm ² application area) per experiment
Group size:	Not mentioned
Skin integrity:	transcutaneous electrical resistance (TER), TER > 5 kΩ
Test substance:	Basic Blue 99
Batch:	RS27988101
Purity:	Content of the main component 43.4 % w/w
Test item:	2 experiments in aqueous solution and formulated material (cream) both at 1.09% w/w Basic Blue C059
Dose volume:	Aqueous solution: 10 µL/cm ² or Formulated solution 10mg/cm ² (109 µg/cm ² of test substance)
Receptor fluid:	25% ethanol in physiologic saline solution
Method of Analysis:	HPLC-UV at 271 nm
GLP:	in compliance
Study period:	12 September 2002 – 16 October 2002

Results

Penetration of Basic Blue 99 through dermatomed pig skin was below the limit of quantitation (LOQ = 0.004 µg/cm²/h penetration rate over 48 h; = 0.2 µg/cm² or 0.18% of dose penetrated at 48h) for both the aqueous Solution and formulation.

Mass balance and Basic Blue 99 distribution

The mean recovery of Basic Blue 99 was between 74% and 78% from the aqueous solution and between 75% and 79% from the formulated material, which was below the target value of 85%.

The majority of the applied Basic Blue 99 was washed from the surface of the skin at 0.5 h (aqueous solution = mean 64.6 %; formulated material = mean 70.5 %). A further 4.37 % and 3.28% of the dose, respectively, was removed from the surface during the terminal wash at 48 hrs.

No Basic Blue 99 was recovered from the tape strips, representing the stratum corneum (total <3.25 % of dose) for both the aqueous solution and formulation, however small amounts (mean 3.12 % and 0.549 %, aqueous solution and formulation, respectively) were recovered from the remaining epidermis/dermis. The standard deviations were respectively for the epidermis/dermis 3.68 and 0.264 for solution and formulation respectively.

Applicants' conclusions

1. The mean recovery of Basic Blue 99 was slightly less than the target value of 85%, at 78% from the aqueous solution and 79% from the formulated material.
2. The vast majority of the Basic Blue 99 was removed after a contact period of 0.5h by simple rinsing the skin surface (64.6% from the aqueous Solution and 70.5% from the formulated material).
3. No Basic Blue 99 was detected as having penetrated through the skin into the receptor fluid from either of the applications (<0.183% of the applied dose).

4. No Basic Blue 99 was extracted from the stratum corneum (<3.25% of the applied dose).
 5. After tape stripping, only 3.12% and 0.549% of applied Basic Blue 99 was found in the remaining epidermis/dermis from the aqueous Solution and formulation, respectively, however this is believed to be contamination from the flange area of the skin under the donor chamber.

6. The maximum systemically available dose (epidermis/dermis plus receptor fluid, including LOQ value as positive values) would be approximately 3.3% for the aqueous Solution and 0.73% from the formulated material. Therefore, these calculated amounts could be used for risk assessments.

7. These data demonstrate that the systemic availability of Basic Blue 99, after dermal exposure to this standard formulation or an aqueous solution of unformulated Basic Blue 99 would be minimal.

According to the Applicant, taking into account that the protocol used is different from the most updated SCCS method recommended from 2004, as only 1 donor, too few test chambers (6 chambers from the donor instead of 8 recommended in the new guideline) and 10 mg/ml of test sample instead of 20 mg/ml recommended for oxidative hair dye (not mentioned for direct hair dye) in the latest guideline, the % of absorption considers the mean + 2SD. To stay at the conservative side, the Applicant recommends using:

- For formulation the mean + 2SD = (0,549 + 0,183) + 2 SD (2 x 0.264) = 1,26% or 1,26% * 109 µg/cm² = 1,37µg/cm²)

- For aqueous solution the mean + 2SD = (3,12+0.183) + 2*3.68*109 = 11.62µg/cm²

Ref.: Ward R. *In vitro* penetration of BASIC BLUE 99 through pig skin from a vehicle and a standard formulation, Study No. JV1718, 2002

SCCS comments

The dermal absorption study has not been considered adequate by the SCCS since it was conducted with an old batch (batch No RS27988101) containing only 43.4 % of the main component, while calculations were wrongly based on 100 %. Therefore, the test substance is not representative with the material specifications outlined in the current Submission. Additional limitations include the use of only one pig, 48 hr exposure, high ethanol content in the receptor fluid, analysis was conducted by a low sensitivity HPLC-UV method and exhibited low recovery of the dose (75 % instead of ≥ 85 %). As such, the study does not provide a reliable basis for the estimation of dermal absorption of the test substance. In the absence of a proper dermal penetration studies a 50% dermal absorption will be used by the SCCS.

Taken from CIR REPORT 2023

Skin Penetration/Dermal Absorption

Wolfram (1984) studied the dermal absorption of [¹⁴C]-labeled Basic Blue 99 in rats and human volunteers. In rats, two animals excreted <0.02% in urine and <0.07% in feces, while one excreted 5.05% in urine and 0.12% in feces; most of the dose was recovered from the skin or patches. In humans, <0.1% of the applied dose was detected in urine, indicating very low systemic absorption.

Ref: Andersen FA (ed.). Final Report on the Safety Assessment of Basic Blue 99. Int J Toxicol. 2007;26(Suppl. 2):51-63.

SCCS comment

The 1984 study does not follow SCCS Notes of Guidance, the rat data are not accepted. The test substance is not representative of the material in the current submission.

3.3.5. Repeated dose toxicity

3.3.5.1. Repeated Dose (14 days) oral toxicity

/

3.3.5.2. Sub-chronic (90 days) toxicity (oral)

Guideline:	OECD 408 (1998)
Species/strain:	SPF bred Wistar rat
Group size:	10 male and 10 female rats
Test substance:	Commercial grade Basic Blue 99 (C059) in aqueous solution
Batch:	74/75
Purity:	Not reported in the study report but 62.8 % (based on HPLC – Dossier submitted to SCCS on April 2013)
Vehicle:	water
Dose levels:	0, 10, 60 and 240 mg/kg bw/day
Dose volume:	10 mL/kg daily administration
Route:	oral
Administration:	daily during 13 weeks
GLP:	In compliance
Study period:	June 2005 – July 2005

Basic Blue 99, in aqueous solution, was administered by oral gavage daily to groups of 10 male and 10 female rats at doses of 10, 60 and 240 mg/kg bw/day for 13 weeks. An additional 5 males and 5 females were treated at the same doses then maintained without treatment for observation in a 4-week recovery period. Controls received the vehicle only. The following investigations were performed: daily observations, body weights, food consumption, ophthalmology, hematology and clinical chemistry, urinalysis, gross pathological examination, organ weight determination and histopathology.

Results

Daily administration of C059 at dose of 10, 60 and 240 mg/kg/day for 13 weeks followed by a 4 week treatment free recovery period resulted in no early mortality, no effect on food and water consumption and body weight development. There was no evidence of eye toxicity.

Effect recorder upon daily and weekly clinical observation were restricted to stained blue urine and feces associated with the intense blue color of the test item

Measurement performed during the functional observed battery revealed no relevant change to grip strength or motor activity.

Clinical investigations revealed only minimal slight changes to few red blood cell parameters, granulocytes and large unstained cells. In addition, lower concentration of glucose, urea and creatinine of poor diagnostic relevance and a minimum slight to slight extent of their impairment are not considered of adverse effect.

A minimal disturbance to the electrolyte (or water) balance in mid and high dose was substantiated by higher sodium, calcium, chloride concentration. In female there was only minimal reversible increase of the chloride concentration.

Change in phosphate concentration was noted on mid and high dose animals (males increase, female decrease) with poor dose relationship. This finding persisting to recovery end in high dose males is considered of equivocal toxicological relevance

At necropsy the absolute and relative mean organ weights were not affected, and macroscopic changes were restricted to bluish discoloration in the stomach at mid and high dose males and high dose females. At the recovery end, accentuated lobular pattern was noted in all high dose males

Microscopically there were no change of toxicological relevance to the morphology of tissue and organs.

Conclusion

Under the condition of this study a NOEL could not be established. In the absence of morphological changes to organs and tissue and minimal and reversible effects on clinical laboratory parameter the NOAEL is defined at 60 mg/kg bw/ day for male and female

According to the Applicant, as the dye content of the test substance is only 62.8 %, the NOAEL value is re-calculated to be 37 mg/kg/bw/day.

Ref.: Bachmann MFDKG. 90-Day Oral (Gavage) Toxicity Study in Wistar Rats Followed by a 4 Week Recovery Period, Report No. R 0600019 (Vol. I and II). 2006

SCCS comment

There is a variation in both the composition and impurity profile between Batch 74/75 used in this study and the new batches. Thus, the toxicological data that are based on Batch 74/75 may not fully correspond to the material specifications outlined in the current Submission

3.3.5.3. Chronic (> 12 months) toxicity

/

3.3.6. Mutagenicity / Genotoxicity

3.3.6.1. Mutagenicity / Genotoxicity *in vitro*

IN VITRO STUDY #1: Bacterial Reverse Mutation test

Guideline: OECD 471 (1997)
 Species/strain: Salmonella typhimurium TA98, TA100, TA102, TA1535 and TA1537
 Replicates: Triplicates in one experiment both in the presence and absence of S9- mix.
 Test substance: C 059
 Solvent: DMSO
 Batch: 74/75
 Purity: Not reported in the study report but 62.8 % (based on HPLC – Dossier submitted to SCCS on April 2013)
 Concentrations: 0, 3.16, 10, 31.6, 100, 316 and 1000 µg/plate without and with metabolic activation (S9 rat liver :2mg/ml)
 GLP: in compliance
 Study period: June 2005

C 059 was investigated for the induction of gene mutations in Salmonella typhimurium (Ames test). Liver S9 fraction from phenobarbital/β-naphthoflavone-induced rats was used as exogenous metabolic activation system. Depending of the bacterial strain precipitate and/or cytotoxicity was observed in higher concentration 316 and 1000µg/plate, which according to the OECD 471 justify the tested concentration range.

The solvent control produced a low number of revertants.

Cytotoxicity was evaluated for 6 concentrations up to a concentration of 1000 µg/plate. Negative and positive controls were in accordance with the OECD guideline, except the failure of the positive control for TA 1537 without S9, which did not impair the result of the study as the functionality of the test system can be demonstrated.

Results

An increase in the number of revertant colonies and a possible dose-effect relationship were noted with and without S9 in the following strains TA 98, TA100, TA 1537.

A dose-effect relationship was also observed in presence of S9.

However, for the strains TA 100 and TA1537 a mutagenic potential of C059 was determined only at concentration where cytotoxicity and / or precipitate could be observed as well.

Conclusion

Under the experimental conditions used C 059, a biological relevant mutagenic potential with and without metabolic activation was determined under the condition of this test.

Ref: Krätzer F. 2005

SCCS comment

A positive result is observed in this study. The test substance, batch 74/75, is not representative with the material specifications outlined in the current Submission.

IN VITRO STUDY #2: Bacterial Reverse Mutation Assay

Guideline :	Guideline : OECD 471 (2020)
Species/strain :	<i>Salmonella typhimurium</i> TA98, TA100, TA102, TA1535 and TA1537
Replicates :	Triplicates in two independent experiments both in the presence and absence of S9.
Test substance :	Basic Blue 99, chemical name Benzenaminium, 3-[(4-amino-6-bromo-5,8-dihydro-1-hydroxy-8-imino-5-oxo-2-naphthalenyl)amino]-N,N,N-trimethyl-, chloride, also identified Arianor Steel Blue 306014, (CAS number 68123-13-7)
Solvent :	water
Batch :	VW080224A
Purity :	Stated as 100% since tested in its entirety (deviation to the recommendation of the GLP, it does not the validity of the study and its GLP compliance) Practices
Concentrations :	80-5000 µg/plate for TA102 strain without S-9 and TA98, TA100 and TA102 strains with S-9, 40-2500 µg/plate for TA98 and TA100 strains without S-9 and TA1535 and TA1537 strains with S-9 and 20-1250 µg/plate for TA1535 and TA1537 strains without S-9
GLP :	in compliance
Study period :	April 2024

Basic Blue was investigated for the induction of gene mutations in 5 strains of *Salmonella typhimurium*, both in the absence and the presence of metabolic activation by a Phenobarbital-5,6 Benzoflavon-induced rat liver post-mitochondrial fraction (S-9). Toxicity was observed in the TA100, TA1535 and TA1537 strains from 500 µg/plate without S-9 and from 1600 µg/plate with S-9 and in the TA98 and TA102 strains from 1600 µg/plate without S-9 and at 5000 µg/plate with S-9. It was observed from 1250 µg/plate in all strains without S-9 and in TA102 strain with S-9, and from 625 µg/plate in TA98, TA100, TA1535 and TA1537 strains with S-9. Precipitates were observed on the test plates following incubation at 5000 µg/plate whatever the strain and the conditions, and from 1250 µg/plate during the second experiment. Negative and positive controls were in accordance with the OECD guideline.

Results

Following treatments of TA98 strain with S-9 resulted in a ≥ 2 -fold increase in revertant numbers. This increase was a significant bell-shaped from 16 µg/plate to 1600 µg/plate (ratio value: 2.9, up to 20.8 at 160 µg/plate) using plate incorporation method in the First Experiment and significant from 80 µg/plate to 1250 µg/plate (ratio value: 2.9 up to 4.0 at 160 µg/plate) using pre-incubation method in the Second Experiment. Following treatments of TA100 strain with S-9 resulted in a ≥ 2 -fold significant increase in revertant numbers at 500 and 1600 µg/plate (ratio value: 2.2) using plate incorporation method in the First Experiment and at 625 µg/plate (ratio value: 2.4) using pre-incubation method in the Second Experiment. Following treatments of TA1537 strain using plate incorporation resulted in a ≥ 3 -fold significant dose-related increase in revertant numbers at 160 and 500 µg/plate (ratio values: 3.3 and 3.8 respectively) with S-9 in the First Experiment, and starting at 50 µg/plate (ratio value: 2.8 statistically significant) and at 300 and 625 µg/plate (ratio values: 3.0 and 5.9 respectively) without S-9 in the Second Experiment.

No other significant increases in revertant numbers were observed in TA98 and TA100 without S-9 or TA102 with and without S-9) or in TA1535 with and without S-9).

Conclusion

Basic Blue 99 induced mutation in histidine-requiring strains TA98, TA100 and TA1537 of *Salmonella typhimurium* when tested under the conditions of this study. These conditions

included treatments using plate incorporation and pre-incubation methods at concentrations up to 5000 µg/plate a toxic precipitating dose in the absence and the presence of a rat liver metabolic activation system (S-9).

Ref: Bellet D., 2024

SCCS comment

A positive result is observed in this study. The study was conducted with a new batch of the test substance.

IN VITRO STUDY # 3: Bacterial Reverse Mutation Assay

Guideline:	(no TG in 1977)
Species/strain:	S typhimurium strains TA1535, TA1537, TA1538
Replicates:	duplicates
Test substance:	1,5-diamino-3-(1'-aminophenylene-3'-trimethyl ammonium chloride)-7-Br- naphthoquinone-4.8
Solvent :	DMSO
Batch :	No info
Purity :	No info
Concentrations :	1, 10, 100 and 1000 µg/plate
GLP:	no
Study period :	1977

With strain TA1535 there was no evidence of mutagenic potential. With strain T1537 there was evidence of mutagenic potential in the absence of metabolic activation, but not on its presence. With strain 1538, there was evidence of mutagenic potential only in the presence of S9. This compound appears to be a frameshift mutagen.

REF: Hossak *et al.* (1977).

SCCS comment

This study suggests mutagenicity. However, the study is of limited reliability due to the use of only 3 of the 5 required strains, insufficient number of concentrations tested, limited reporting on methodology. The study was also conducted with an older batch of the test substance, which is not representative of the material specifications outlined in the current Submission.

IN VITRO STUDY #4: Bacterial Reverse Mutation Assay

Guideline	-
Species/strain:	S typhimurium strains TA98, TA100, TA1535, TA1537, TA1538
Replicates:	-
Test substance:	Arianor Steel Blue KS 3993
Solvent:	-
Batch:	C.I. number 56059
Purity:	No info
Concentrations:	4, 20, 100, 500 and 2500 µg/plate
GLP:	no
Study period:	1985

The sample induced reverse mutations in the absence of S9 in the tester strain TA100 -and in the presence of S9 in the strains TA98, TA100, TA1537, TA1538.

The sample Arianor Steel blue K 3993 showed mutagenic activity *in vitro*.

Ref.: Wallat S. (1985)

SCCS comments

Although the full report is in German, the English summary indicates a valid study showing mutagenicity in strains TA102 and TA100 with metabolic activation (+S9), as well as in TA1537 and TA98 both with and without S9. The test substance, batch 56059, is not representative of the material specifications outlined in the current Submission.

IN VITRO STUDY # 5: Bacterial Reverse Mutation Assay

Guideline :	OECD 471 (1997)
Species/strain :	<i>S typhimurium</i> strains TA1535, TA1537, TA98 TA100, and TA102
Replicates :	triplicates
Test substance :	Arianor Steel Blue
Solvent :	Deionised water
Batch :	RS 27988101 (Henkel Code: SAT 000939)
Purity :	>55 Area % (HPLC)
Concentrations :	Maximum dose of 4 to 2500 µg/plate TA98 and TA100: - 3, 10, 33, 100, 333, and 1000 µg/plate +/-S9 TA1535, TA1537, and TA102: - 3, 10, 33, 100, 333, and 666 µg/plate -S9 - 33, 100, 333, 666, 1000, and 2500 µg/plate +S9
GLP:	Yes
Study period :	October 2000

A substantial increase in revertant colonies was observed following treatment with Arianor Steel Blue in strains TA1537 and TA98 with and without S9 and in strains TA102 and TA100 with S9. Therefore, Arianor Steel Blue is considered to be mutagenic (with and without metabolic activation in this *S typhimurium* reverse mutation assay.

REF: Cytotest Cell Research GmbH (2000).

SCCS comments

The study is considered valid and positive. The test substance, batch RS 27988101, is not representative with the material specifications outlined in the current Submission.

IN VITRO STUDY # 6: In vitro Mammalian Cell Gene Mutation Test (tk-locus)

Guideline :	OECD 476 (2000)
Species/strain:	L5178Y Mouse lymphoma cells
Replicates:	3 independent experiments, using 2 parallel cultures each
Test substance:	C 059
Solvent:	deionised water
Batch:	74/75
Purity:	Not reported in the study report but 62.8 % (based on HPLC – Dossier submitted to SCCS on April 2013)

Concentrations: Experiment I: 0, 1.3, 2.5, 5.0, 10, 15, 20, 30 and 40 µg/mL, **4 hours treatment with and without S9-mix**
Experiment II: 0, 1.3, 2.5, 5.0, 10, 15, 20, 30 and 40 µg/mL, **24 hours treatment, without S9-mix.**
An additional confirmatory experiment III: 0, 5.0, 7.5, 10, 12.5, 15.0 and 17.5 µg/mL, 4 hours **and 24 hours treatment with and without S9-mix.**

GLP: in compliance

Study period: June 2005 – December 2005

C 059 was assayed for gene mutations at the *tk* locus of mouse lymphoma cells both in the absence and presence of S9 metabolic activation. Liver S9 fraction from phenobarbital/β-naphthoflavone-induced rats was used as exogenous metabolic activation system. Test concentrations were based on the results of a pre-test on toxicity measuring relative suspension growth.

Results

No substantial and reproducible dose dependent increase in mutant colony numbers was observed in the main experiments. No relevant shift of the ratio of small versus large colonies was observed up to the maximal concentration of the test item.

Appropriate reference mutagens were used as positive controls and showed a distinct increase in induced mutant colonies, indicating that the tests were sensitive and valid.

Conclusion

The authors concluded that under the experimental conditions reported the test item did not induce mutations in the mouse lymphoma thymidine kinase locus assay using the cell line L5178Y in the absence and presence of metabolic activation.

C059 is considered to be non-mutagenic in this mouse lymphoma assay.

Ref.: Wollny HE, 2005

SCCS comment

A positive result is observed in Experiment I after 4h of exposure, without S9, evident at the two highest concentrations of 15 and 20 µg/mL, but 20 µg/mL was too toxic to be considered in Exp I and II. At 24h, the ExpII is unreliable and considering the Exp III, an inconsistent increase is also observed at 4h and 24h at the concentration of 15 µg/mL. The test system shows unacceptable fluctuations in response between cultures and experiments. Overall, the results of this study suggest equivocal mutagenic effects of Basic Blue 99. The test substance, batch 74/75, is not representative with the material specifications outlined in the current Submission.

IN VITRO STUDY # 7: *In vitro* Mammalian Cell Gene Mutation Test (hprt locus)

Guideline : OECD 476 (2016) and UKEMS guidelines
 Cells : L5178Y Mouse lymphoma cells
 Replicates : 2 replicate cultures
 Test substance : Basic Blue 99 (CAS number 68123-13-7), also known as
 ARIANOR STEEL BLUE 306014
 Solvent : purified water
 Batch : VW080224A
 Purity : 69.5 %
 Concentrations : 0, 0.75, 1.5, 3, 7, 8, 9, 10 and 11 µg/mL, without S9-mix
 0, 1.5, 3, 6, 10, 12.5, 25 and 30 µg/mL, with S9-mix
 GLP : yes (audited report available)
 Study period : March 2024 – June 2024

Basic Blue 99 was assayed for the ability to induce mutation at the hprt locus in mouse lymphoma cells using. The study consisted of two cytotoxicity Range-Finder Experiments followed by a Mutation Experiment, each conducted in the absence and presence of metabolic activation by a β Naphthoflavone/Phenobarbital-induced rat liver post-mitochondrial fraction (S-9). A 3 hour treatment incubation period was used for each experiment. Twelve concentrations, ranging from 0.75 to 25 µg/mL, were tested in the absence of S-9 and thirteen concentrations ranging from 1.5 to 50 µg/mL, were tested in the presence of S-9. Seven days after treatment, the highest concentrations analysed to determine viability and 6TG resistance were 11 µg/mL in the absence of S-9 and 30 µg/mL in the presence of S-9, which gave 16% and 13% RS (relative survival), respectively.

Vehicle and positive controls were included in the Mutation Experiment, with clear increases mutant frequencies for the positive controls.

Results

When tested up to toxic concentrations for 3 hours in the absence and presence of S-9, no statistically significant increases in mean MF, compared to the mean vehicle control values, were observed.

Conclusion

It is concluded that Basic Blue 99 was not mutagenic at the hprt locus in mouse lymphoma L5178Y cells when tested up to toxic concentrations for 3 hours in the absence and presence of a rat liver metabolic activation system (S-9), under the experimental conditions described.

Ref: Nisar, 2024

SCCS comment

A negative result is observed in this study up to 11 µg/mL. The conclusion is sound, although the concentrations tested are lower than *Ref 14. Wollny et al.* due to excessive toxicity. Why this excessive toxicity, at 3h exposure, is not observed in the other study done in the same cell type remains unexplained. Test item purity differences may have been the cause, and this negative result therefore does not overrule the previous findings. The study was conducted with a new batch of the test substance.

IN VITRO STUDY # 8: *In vitro* Mammalian Cell Gene Mutation Test (Hprt locus)

Guideline : OECD 476 (2016)
 Cells : V-79 Chinese hamster cells (Hprt locus)
 Replicates : 2 replicate cultures
 Test substance : Arianor Steel Blue
 Solvent : DMSO
 Batch : KS 3993, Williams
 Purity : dye content: 60.7 %
 Concentrations : 24 hours, -S9:
 0.1, 0.3, 0.5, 1.0, 2.5, 3.0, 5.0, or 10.0 µg/ml (4 separate experiments)
 2 hours, + S9:
 3, 10, 25, 30, 50, 100, 125, 150, or 250 µg/ml
 GLP : yes
 Study period : May – September 1991

In the concentration range investigated, Arianor Steel Blue (Basic Blue 99) did not show mutagenic activity both with and without the presence of metabolic activation system.

Ref.: Michalke H., (1991).

SCCS comments

Negative study with limited reliability: exposure for 2 h +S9, slight (not statistically significant nor concentration dependent) increases in MF at some concentration points (especially -S9 in Exp 1 but not confirmed in repeated experiments) observed, negative historical control range not provided. The test substance, batch KS 3993, is not representative of the material specifications outlined in the current Submission.

IN VITRO STUDY # 9: *In vitro* Micronucleus Test

Guideline : OECD 487 (2004)
 Cells: V79 cells (MZ cell line)
 Replicates: (duplicates)
 Test substance : C 059
 Solvent: DMEM (Dubelco Modified Eagle Medium)
 Batch : 74/75
 Purity: Not reported in the study report but 62.8 % (based on HPLC – Dossier submitted to SCCS on April 2013)
 Concentrations: 0, 1.56, 3.13 and 6.25 µg/ml without S9-mix (decrease in number of cell attached at 25 and 50 µg/ml),
 0, 37.5, 75 and 150 µg/ml with S9-mix (decrease in number of cell attached at 150 and 300 µg/ml;
 Presence of precipitation observed at 75, 150 and 300µg/ml),
 4 h treatment; harvest time 24 hours after the beginning of treatment
 GLP: in compliance
 Study period: March 2005 – September 2005

C 059 has been investigated in the absence and presence of metabolic activation for the induction of micronuclei in V79 cells. Liver S9 fraction from phenobarbital/ β naphthoflavone-induced rats was used as exogenous metabolic activation system. A pretest on cell growth inhibition (MTT assay) with 4 h treatment was performed in order to determine the toxicity of C 059, the solubility during exposure and thus the test concentrations for the main micronucleus test.

The highest concentration should produce clear toxicity with reduced cell growth. The treatment period in the main test was 4 h without and with S9-mix. Harvest time was 24 hours after the beginning of culture. In parallel to the micronucleus test, for assessment of cytotoxicity a MTT test was carried out.

Negative and positive controls were in accordance with the draft guideline.

Results

The test item C059 was assessed for its potential to induce micronuclei in V79 cells *in vitro* both in the presence and in the absence of a metabolizing system (S9).

The exposure was performed without and with metabolic activation (2 mg/ml rat liver S9) for 4 h with the test.

At the end of incubation without S9 a decrease in the number of attached cells at concentrations of 25 and 50 μ g/ml was noticed. Similar effects were seen at the end of incubation with S9 at concentrations of 150 and 300 μ g/ml.

Due to the cytotoxicity, the highest concentration tested was 6.25 μ g/ml for incubations without S9 and 150 μ g/ml for incubations with S9.

The cytotoxicity for these doses was found to be 48.6% and 15.0 %, respectively.

Precipitation was observed at test item concentrations of 300, 150 and 75 μ g/ml only in presence of metabolic activation.

C059 induced a clear increase in the number of micronuclei both with and without metabolic treatment. Furthermore, a dose-effect relationship was observed for incubations with S9.

Conclusion

Thus, under the conditions of this study a micronuclei inducing potential was detected for the test item C059.

Ref: Krätzer F., 2005

SCCS comment

A positive result is observed in this study. Over 5-fold increase in MN frequency at 6.25 μ g/mL without S9 and >3-fold increase at 150 μ g/mL with S9 were observed. The test substance, batch 74/75, is not representative with the material specifications outlined in the current Submission.

IN VITRO STUDY #10: *In vitro* Chromosomal aberrations test

Guideline :	OECD Tg 473 (1998)
Cells:	V-79 Chinese hamster cells
Replicates:	duplicates
Test substance :	Arianor Steel Blue
Solvent:	Deionised water (max. 10% total culture medium)
Batch :	RS 27988101 (Henkel Code: SAT 000939)
Purity:	>55 Area % (HPLC)
Concentrations:	4 h exposure 0, 0.5, 0.8, and 1.5 μ g/mL without S9 and 15, 30, and 45 μ g/mL with S9. The chromosomes were prepared 18 h after the start of treatment.
GLP:	Yes
Study period:	November 2000- January 2001

Without S9 there were significant increases in the number of cells with structural chromosomal aberrations after treatment with 0.5, 0.8, and 1.5 µg/mL Basic Blue 99 (with increases of 4.0%, 14.5%, and 9.5%, respectively).

With S9: significant increase in the number of cells with structural chromosomal aberrations after treatment with 45 µg/mL Basic Blue 99 (with an increase of 12.0%).

Ref.: Cytotest Cell Research GMBH (2001).

SCCS comment

Positive study with limited reliability: 200 metaphases scored, indication of unacceptable cytotoxicity at 1.5 µg/mL –S9-mix (cell number count). The test substance, RS 27988101, is not representative with the material specifications outlined in the current Submission.

IN VITRO STUDY #11: *In vitro* Unscheduled DNA synthesis

Guideline :	OECD TG 486 no longer in place
Cells:	Wistar rat hepatocytes
Replicates:	6
Test substance :	Arianor Steel Blue
Solvent:	L-15 medium
Batch :	KS 3993
Purity:	composed mixture, 60.7% dye content
Concentrations:	Freshly isolated hepatocytes exposed for 3 hours to: 1, 3.33, 10, 33.33 and 100.00 µg/mL
GLP:	Yes
Study period:	August 1988

There was no indication of a distinct dose-related increase in the incorporation of radioactivity, therefore the test article did not induce DNA repair synthesis in hepatocytes.

Ref.: Timm A. (1988)

SCCS comments

A negative result is observed in this study. OECD does no longer have an *in vivo* UDS guideline for mammalian liver cells (Test No. 486). The test substance, KS 3993, is not representative of the material specifications outlined in the current Submission.

3.3.6.2. Mutagenicity / Genotoxicity *in vivo*

IN VIVO STUDY #1: Mammalian Erythrocyte Micronucleus Test

Guideline:	OECD 474 (1997)
Species/strain:	NMRI
Replicates:	10 mice (5 males / 5 Females) per test group
Test substance:	C 059
Batch:	74/75
Purity:	Not reported in the study report but 62.8 % (based on HPLC – Dossier submitted to SCCS on April 2013)
Dose levels:	0, 1.25, 2.5 and 5 mg/kg bw
Sampling time:	24 hours for all doses and 48 hours for the highest doses
Route:	i.p.
Vehicle:	deionised water

Sacrifice times: 24 h after treatment for all concentrations, 48 h for the highest dose only
 GLP: In compliance
 Study period: September – December 2005

C 059 has been investigated for the induction of micronuclei in bone marrow cells of mice. Test concentrations were based in a pre-experiment on acute toxicity at various intervals of 1, 2-4, 6, 24, 30 and 48 h after start of treatment. Doses tested in pre-experiment were 100, 50, 10, 1, 7.5 and 5mg/kg.bw. In the main experiment mice were exposed i.p. to 0, 1.25, 2.5 and 5 mg/kg bw. Bone marrow cells were collected 24 h or 48 h (highest dose only) after dosing. Toxicity and thus exposure of the target cells was determined by measuring the ratio between polychromatic and total erythrocytes (PCE/TE).

The mice of the high dose group (5mg/kg.bw) were examined for acute toxic symptoms at intervals of around 1, 2-4, 6 24 and 48h after treatment. Bone marrow preparations were stained with May-Grünwald/Giemsa and examined microscopically for the PCE/TE ratio and micronuclei. Negative and positive controls were in accordance with the OECD guideline.

Results

In the a pre-experiment on acute toxicity with exposure up to 7.5 mg/kg.bw C059, reduction of spontaneous activity, abdominal position, ruffled fur, apathy and death were recorded. 2 animals (1 male, 1 female) died.

In the main experiment only reduction of spontaneous, abdominal position and ruffled fur were found at the highest doses. Treatment with C 059 did not result in a decreased PCE/TCE ratios compared to the untreated controls indicating that C 059 had no cytotoxic properties in the bone marrow. However, the acute toxicity and particularly the blue colored urine, indicate systemic availability of C 059.

Conclusion

Under the experimental conditions used C 059 did not induce a biologically relevant increase in the number of micronucleated PCEs in bone marrow cells of treated mice and consequently, C 059 is not regarded clastogenic and/or aneugenic in bone marrow cells of mice.

Ref: Honarvar N. (2005)

SCCS comment

A valid negative result is observed in this study. The test substance, 74/75, is not representative of the material specifications outlined in the current Submission.

IN VIVO STUDY #2: Mammalian Erythrocyte Micronucleus Test

Guideline: OECD 474 (1997)
 Species/strain: Mice; 6 animals per group
 Replicates: ?
 Test substance: Arianor Steel Blue
 Batch: ?
 Purity: no info
 Dose levels: 1500 mg/kg
 Sampling time: 24, 48 and 72 hours
 Route: Gavage
 Vehicle: NaCl 0.9%
 Sacrifice times: 24h (male and female animals), 48h (male and female animals) and 72h (female animals only) after exposure
 GLP: (mentioned in German)
 Study period: August 1987

The test article induced high rates of mortality in the male treatment groups of the 24h (3/6) and 72h (4/6) observation periods. No mortality was observed in the male or female 48 hr group.

In the remaining groups, there was no indication of a mutagenic action of the test article *in vivo* and no indication of a delayed cell proliferation.

It was concluded that the test item failed to show any evidence of mutagenic potential when administered orally.

Ref.: Banduhn N. (1987). Aranior Steel Blue. Prüfung auf Mutagenität im Mikrokern-Test *in vivo*. Unpublished data submitted by Henkel, in German with English summary. 27 pages.

SCCS comments

The test was negative, tested at only one high dose which induced mortality; a lower dose should have been tested to validate the absence of genotoxicity. The test substance, used for this study conducted back in 1987, is not representative of the material specifications outlined in the current Submission.

IN VIVO STUDY #3: Mammalian Erythrocyte Micronucleus Test

Guideline :	OECD 474 (1997)
Species/strain:	Mouse NMRI (SPF)
Replicates:	10 animals (5 males, 5 females)
Test substance :	SAT 000939
Batch :	RS 27988101
Purity:	No info
Dose levels:	0.2, 1.0, and 5.0 mg/kg bw
Sampling time:	24 hours post-administration and also at 48 h for highest dose
Route:	Intraperitoneal injection
Vehicle:	Bidestilled water
Sacrifice times:	24 hours post-administration and also at 48 h for highest dose
GLP:	Yes
Study period:	September-November 2001

No statistically significant or biologically relevant increase in the frequency of detected micronuclei at any preparation interval or dose level.

In pre-experiment for toxicity, death occurred in both of 2 animals treated at 10 mg/kg.

In the main experiment 1 animal died at 5 mg/kg 3 hrs after the treatment (cause of death could not be established).

Ref: RCC Ltd. (2001b).

SCCS comments

Negative results were observed. However, the general toxic symptoms in animals treated with the test item indicate its systemic distribution. The test substance, RS 27988101, is not representative of the material specifications outlined in the current Submission.

IN VIVO STUDY #4: Unscheduled DNA Synthesis Test with Mammalian Liver Cells *In Vivo / In vitro*

Guideline :	OECD 486 (1997)
Species/strain:	rat, Wistar HanIbm WIST (SPF)
Replicates:	32 (males) : Four male rats for each of the 8 test groups
Test substance :	C059 Basic Blue 99
Batch :	RS27988101
Purity:	Not reported in the study report but 50.2% (based on area – Dossier submitted to SCCS on April 2013)
Dose levels:	0, 1000, 2000 mg/kg bw Dosing volume: 10 mL per kg bw
Sampling time:	
Route:	Oral, single dose
Vehicle:	De-ionised water for injection
Control:	dimethylhydrazinedihydrochloride(DMH), 40 mg/kg.bw, 2-hour treatment group: 2-acetylaminofluorene (2AAF), 100 mg/kg bw, 16-hour treatment group
Treatment	2 and 16h treatment
Sacrifice times:	2 and 16h
GLP:	In compliance
Study period:	June 2003 – September 2003

An *in vivo/in vitro* unscheduled DNA synthesis assay with rat hepatocytes was conducted to examine the DNA damaging potential of C059, Basic Blue 99.

As a result of the pre-experiment with 2 males and 2 females rats receiving a single dose of 2000 mg/kg bw, 1 death was observed after one hour, but considered as not test item related. Then 2000mg/kg.bw, the recommended dose by OECD guideline, was considered as suitable. Therefore, two oral doses were used to compare a 1000 & 2000 mg/kg dose

Hepatocytes were isolated by the collagenase perfusion method at 2 or 16 hours after administration and cultured. Thereafter, the nuclear grain count and cytoplasm grain count were determined.

Results

The mean net nuclear grain (NNG) counts of the C059 Basic Blue 99 treatment groups were comparable to those of the negative control group. The incidence of cells in repair (UDS positive) in the treatment groups was also comparable to that of the negative controls group, showing no remarkable trend of increase.

Neither the nuclear grains nor the resulting net grains were distinctly enhanced due to the *in vivo* treatment of the animals with the test item for 2 hours or 16 hours respectively.

In two positive control groups, i.e., in a group treated with dimethylhydrazine dihydrochloride (DMH), 40 mg/kg.bw, 2-hour treatment and in a group treated with 2-acetylaminofluorene (2AAF: 100 mg/kg dose) for 16 hours, the hepatocytes of all rats showed large counts of NNG and high DNA repair cell appearance frequencies, indicating a positive reaction.

Conclusion

It was concluded that C059 did not induce DNA damage leading to increased repair synthesis in the hepatocytes of the treated rats. Therefore, Basic Blue C059 is considered to be non-effective in this *in vivo / in vitro* UDS test system.

Ref.: Honarvar N. (2003)

SCCS comment

A negative result was observed in this study; however, it is considered not informative. According to the SCCS Notes of Guidance (2022), the Unscheduled DNA Synthesis (UDS) test *in vivo* using mammalian liver cells is no longer recommended. The test substance,

RS27988101, is not representative with the material specifications outlined in the current Submission.

IN VIVO* STUDY #5: Unscheduled DNA Synthesis Test with Mammalian Liver Cells *In Vivo

Guideline :	OECD 486 (1997)
Species/strain:	Wistar rat hepatocytes
Replicates:	3
Test substance :	C 059
Batch :	R96003970 (SAT 010561)
Purity:	>99%
Dose levels:	250, 500, or 1000 mg/kg
Sampling time:	Hepatocytes isolated 2-4 h or 12-16 h after the dosing.
Route:	Gavage
Vehicle:	Physiological saline
Control:	saline
Treatment	Orally
Sacrifice times:	2-4 h or 12-16 h after the dosing
GLP:	Yes
Study period:	January 2002

No induction of DNA repair in hepatocytes isolated 2-4 h or 12-16 h after dosing with Basic Blue 99 at doses up to 1000 mg/kg.

Ref.: Meerts I. (2002).

SCCS comment

The result was negative but not considered informative. According to the SCCS Notes of Guidance (2022), the Unscheduled DNA Synthesis (UDS) test *in vivo* using mammalian liver cells is no longer recommended. The test substance, C059, is not representative with the material specifications outlined in the current Submission.

Applicants' overall conclusion for mutagenicity/genotoxicity

Basic Blue 99 did induce gene mutations in the gene mutation tests in bacteria (with 2 different batches: 74/75 and VW080224A. However, with 2 different batches, no gene mutations are induced in mammalian cells, neither in the *in vitro* mouse Lymphoma Assay (hprt locus) (VW080224A), neither in mouse Lymphoma Assay (TK+/-) (batch 74/75). The induction of chromosome aberrations by Basic Blue 99 was studied in an *in vitro* on V79 cells and *in vivo* micronucleus test mice bone marrow cells. While exposure did result in an increase of cells with micronuclei in the *in vitro* test, this positive result is invalidated by the *in vivo* micronucleus negative test. Basic Blue 99 can be considered as negative for clastogenic or aneugenic potential in bone marrow cells in mice. In the UDS test, Basic Blue 99 did not induce DNA damage leading to increased repair synthesis in the hepatocytes of the treated rats. Overall, the genotoxicity of Basic Blue 99 is sufficiently investigated for the three endpoints of genotoxicity: gene mutations, chromosome aberrations and aneuploidy and can be considered not genotoxic.

SCCS overall comment on mutagenicity and genotoxicity

Notable differences in the purity/impurity profile of the test item are evident throughout the available data – both submitted by the Applicant in the current and previous submissions, as well as from the studies identified above (see physicochemical section). These differences

may have contributed to the inconsistent findings that do not allow drawing a clear conclusion on genotoxicity. Genotoxicity studies using older batches are not considered relevant by the SCCS because the previously used test materials are not representative of the current batch specifications outlined in this submission. The two studies on the current batch (VW080224A) also provide partial and insufficient information. Therefore, concerns remain about the genotoxic potential of C59 Basic Blue 99.

3.3.7. Carcinogenicity

As genotoxicity of Basic Blue 99 cannot be excluded, the SCCS considers that the concerns for carcinogenicity cannot be ruled out.

3.3.8. Reproductive toxicity

3.3.8.1. Two generation reproduction toxicity

/

3.3.8.2. Other data on fertility and reproduction toxicity

/

3.3.8.3. Developmental Toxicity

Guideline: OECD 414 (2001)
 Species/strain: Rat, Han Wistar, (SPF)
 Group size: 25 per dose
 Test substance: C059
 Batch: 74/75
 Purity: Not reported in the study report but 62.8 % (based on HPLC – Dossier submitted to SCCS on April 2013)
 Dose: 0, 60, 200, 500 mg/kg bw/day
 Solvent: Distilled water
 Route: Oral gavage
 Exposure: Gestation Day (GD) 6 – 20 inclusive
 GLP: in compliance
 Date: June – July 2005

Gestation Day (GD) 0 or mating day was considered as either the presence of sperm in the vaginal smear or by the presence of a copulation plug. Doses were based on the results of the previously performed studies (Henkel Report TBD 900354 (on Reprotoxicity study with a single dose of 50mg/kg.bw) and TBD 860009 (on 90-day study with dose of 20, 60, 180 and 360mg/kg.bw/day). The animals were observed at least twice daily; clinical signs and deaths were recorded. Body weight gain was recorded daily GD 0 until day 21 post coitum. The dams were killed on GD 21 for postmortem. The number of alive and dead fetuses, their distribution and site in the uterus, early and late resorption, implantation and number of corpora lutea was determined. The weight of the fetuses, gravid uteri, uteri without fetuses, placenta and the sex of fetuses were recorded. Approximately one-half of the fetuses were selected at random and examined for visceral alterations. The remaining fetuses were examined for skeletal malformations, variations and retardation of the normal organogenesis after appropriate staining.

Results

Occurrence of black feces, due to the content of test item, was observed in all animals of group 3 and 4 from day 7 post coitum onwards. The bedding material was bluishly discolored in all groups treated with the test item most likely due to the excreted test item in the urine. One Dam from group 4 was found dead in the day morning of Day 11 post coitum. However, its death was considered to be not related to the treatment with the test item. No test item related effect was noted for Food Consumption and Body Weights. The relevant reproduction parameters (number of post implantation loss, number of live and dead fetuses) were similar in all groups and not influenced by treatment with the test item.

Necropsy findings were related for all animals to the bluish discolored of the organs, (stomach and intestine wall, also kidney and placenta show a bluish discoloration), which was considered to be due to the color of the test item and considered to be not an adverse effect. Referring to the fetus, the sex ratio was close to 50/50 and not influenced by the test item. Statistically significantly increased mean body weights were calculated in group 4 male and female fetuses. External examination did not reveal any item related findings Visceral examination did not reveal any item related findings Referring to Skeletal and cartilage examination: no test item abnormalities were observed. Only both individual and litter basis, statistically significant decreased incidences of non-ossified talus and non-ossified cervical vertebral body 1 were found in group 4.

Conclusion

A NOAEL for embryo/fetal toxicity was at the dose level of 200 mg/kg.bw/day. The NOEL for embryo/fetal toxicity was at the dose of 60 mg/kg.bw/day. The study did not reveal any teratological or embryo lethal effect up to the high dose level of 500 mg/kg body weight.

According to the Applicant, as the dye content of the test substance is only 62.8%, the NOAEL value is re-calculated to 125.8 mg/kg bw/day.

Ref.: Gerspach, RFD (2006)

SCCS comment

There is a variation in both the composition and impurity profile between Batch 74/75 used in this study and the new batches. Thus, the toxicological data that are based on Batch 74/75 may not fully correspond to the material specifications outlined in the current Submission

3.3.9. Toxicokinetics

3.3.9.1. Toxicokinetics in laboratory animals

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3.3.9.2. Toxicokinetics in humans

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3.3.10. Photo-induced toxicity

3.3.10.1. Phototoxicity / photo-irritation and photosensitisation

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3.3.10.2. Photomutagenicity / photoclastogenicity

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3.3.11. Human data**SCCS comment**

Notwithstanding the uncertainty concerning the actual composition of BB99, which may have varied in the course of time, several clinical reports indicate that it is both an immediate-type allergen and a delayed-type contact sensitiser.

As early as 1990, the first case of allergic contact dermatitis to BB99 from a semi-permanent hair dyeing products has been reported, with a positive patch test to BB99. A striking feature was temporary hair loss confined to the area of application and dermatitis (de Groot and Weyland 1990). Allergic contact dermatitis was reported in five Japanese patients; all had a positive patch test to BB99 1% in petrolatum (Nishioka 2025). Severe systemic allergic dermatitis following application of a hair dye was reported, with a positive patch-test to BB99 1% pet (Iwasaki 2024).

Several cases of contact urticaria (syndrome) have been reported following the use of hair dye products containing BB99, with positive skin tests to BB99, mainly, but not only, from Japan (Jagtman 1996, Washio 2017, Vanden Broeke 2014, Numata 2025)

3.3.12. Special investigations

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3.3.13. Safety evaluation (including calculation of the MoS)

According to the Applicant, the margins of safety calculation is based on the dermal absorption of test substance, considering the absorbable substance. The dermal absorption of Basic Blue 99 was estimated to be 1.37µg/cm² (mean + 2SD).

Two conditions are considered: non-oxidative conditions, with Basic Blue 99 used in a formulation, and non-oxidative conditions with Basic Blue 99 in aqueous solution. In both cases, according to the Applicant, the MOS is above 100.

BASIC BLUE 99
(non-oxidative conditions with formulation)

Absorption through the skin	$A = (0.549 + 0.183) * 109$	= 0.8 µg/cm ²
Absorption + 2SD		
$A' = A + 2SD =$	$(0.549 + 0.183 + 2 \times 0.264) * 109$	= 1.37 µg/cm ²
Skin Area surface	SAS	= 580 cm ²
Dermal absorption per treatment	$SAS \times A' \times 0.001^*$	= 0.797 mg
Typical body weight of human		= 60 kg
Systemic exposure dose (SED)	$SAS \times A \times 0.001/60$	= 0.0133 mg/kg bw/d
No observed adverse effect level (90-day, oral, rat recalculated)	NOAEL	= 37 mg/kg bw/d
50% bioavailability **		= 18 mg/kg bw/d
Margin of Safety	NOAEL / SED	= 1353

BASIC BLUE 99
(non-oxidative conditions with aqueous solution)

Absorption through the skin	$A = (3.12 + 0.183) * 109$	= 3.6 µg/cm ²
Absorption + 2SD		
$A' = A + 2SD =$	$(3.12 + 0.183 + 2 \times 3.68) * 109$	= 11.62 µg/cm ²
Skin Area surface	SAS	= 580 cm ²
Dermal absorption per treatment	$SAS \times A' \times 0.001^*$	= 6.74 mg
Typical body weight of human		= 60 kg
Systemic exposure dose (SED)	$SAS \times A \times 0.001/60$	= 0.1123 mg/kg bw/d
No observed adverse effect level (90-day, oral, rat recalculated)	NOAEL	= 37 mg/kg bw/d
50% bioavailability **		= 18 mg/kg bw/d
Margin of Safety	NOAEL / SED	= 160

*conversion µg to mg

** adjusted NOAEL for bioavailability

SCCS comment

Based on the currently available information, it is not possible for the SCCS to conclude on the safety of Basic Blue 99 because the toxicological data that are based on Batch 74/75 may not fully correspond to the material specifications outlined in the current submission. Moreover, the concerns regarding the genotoxic potential remain. Hence, calculation of MoS has not been performed by the SCCS.

3.3.14. Discussion

Physico-chemical properties

The SCCS agrees that the variability in the composition of the test substance has been reduced in the batches prepared by the new manufacturing procedure, and that the content of the five main constituents has been stabilized to greater than 80% in the new batches. However, there is a variation (tenfold difference) in the composition of the second main constituent between the new batches and the old batches. In particular, the second constituent is ~30% in the new batches compared to ~3% in the old batches used for toxicology studies.

The SCCS notes that there is also variation in the impurity profiles between the new batches and Batch 74/75, which was used in previous toxicity studies. Some impurities have significantly increased in the new batches. Thus, the toxicological data that are based on Batch 74/75 or similar old batches (RS27988101) may not fully correspond to the material specifications outlined in the current Submission. Based on the submitted data, for the new batches, the average sum of impurities with potential toxicological impact is 2.17% and the average sum of unidentified impurities is 1.99% in the new batches. In total, this corresponds to approximately 4.16% of impurities for which a TTC approach would be applicable.

The SCCS considers a purity of only 80 % to be too low for the safety evaluation of a synthetic compound, particularly when the test material shows marked differences from the batches used in toxicological studies.

The *in silico* assessment of the toxicity for the impurities lacks clarity, key information, and a scientifically sound methodology. In addition, there is no clear mapping between the impurities reported in the dossier and the chemical structures used in the *in silico* studies. The SCCS considers that the Applicant's claim of reduced toxicological concern for impurities present in the new batches is not substantiated by the *in silico* studies submitted.

The SCCS notes that stability studies of the test substance under the conditions of use must be performed by using the same HPLC-PDA method (at λ_{max}) used to analyse impurities and full reports should be submitted.

General toxicity

There is a variation in both the composition and impurity profile between the new batches and the Batch 74/75 used in the submitted toxicological studies. Thus, the toxicological data that are based on Batch 74/75 or other old batches may not fully correspond to the material specifications outlined in the current Submission and cannot be used to derive a NOAEL without proper *in silico* justification.

Irritation/sensitisation

Acknowledging the uncertainty regarding the exact nature of the Basic Blue 99 batch that caused (partly severe) immediate-type hypersensitivity reactions, the SCCS considers this an alert that such a risk may also be associated with the current new batch under consideration. Basic Blue 99 should therefore be regarded as a skin sensitiser, despite the fact that the available supporting studies were conducted with older batches.

Dermal absorption

The submitted dermal absorption study cannot be considered for the safety evaluation due to use of a non-representative batch and major methodological shortcomings. In the absence of proper dermal absorption studies conducted with a batch representative of the marketed material, a default value of 50% dermal absorption is applied, in line with SCCS Notes of Guidance.

Mutagenicity

Due to differences in purity and impurity profiles across batches, the genotoxicity studies on older batches are not considered relevant. The limited data available on the new batches are insufficient, and concerns regarding genotoxic potential remain.

Carcinogenicity

As genotoxicity of Basic Blue 99 cannot be excluded, the SCCS considers that the concerns for carcinogenicity cannot be ruled out.

Toxicokinetics

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Human data

Several clinical case reports indicate that Basic Blue 99 can act as both an immediate-type allergen and a delayed-type contact sensitiser. Reported reactions include allergic contact dermatitis, contact urticaria syndrome, and in some cases, severe systemic allergic responses.

4. CONCLUSION

1. *In light of the data provided, does the SCCS consider Basic Blue 99 safe, when used as an ingredient at 1 % in non-oxidative hair dye formulations?*

Having evaluated all the available data, the SCCS does not consider the use of Basic Blue 99 safe for use as an ingredient in non-oxidative hair dye formulations because the collective evidence indicates a potential for genotoxicity.

2. *Does the SCCS have any further scientific concerns with regard to the use of Basic Blue 99 in cosmetic products?*

Although there is an uncertainty regarding the exact composition of Basic Blue 99 batch that caused (partly severe) immediate-type hypersensitivity reactions, the SCCS considers this an alert for such a risk with the current batch specifications under consideration. The SCCS regards Basic Blue 99 a skin sensitiser.

5. MINORITY OPINION

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7. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – Appendix 15 - from page 158

8. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – Appendix 15 - from page 158
